Adapting Multi-Agent LLM Equity Research to Indian Markets: Data Integrity, Intraday Planning, Adversarial Verification and Regulatory Readiness

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ABSTRACT

TradingAgents is an open-source framework in which large-language-model (LLM) agents take the roles of an equity research and trading team. This report describes an independent fork that adapts it to the National Stock Exchange of India (NSE) and BSE, and examines what it would take to offer such a system in India. The fork adds Indian symbol handling, the 2026 exchange calendars and session rules applied by date, rupee conventions, configurable Indian transaction charges and tax rules, Indian benchmarks and news windows in Indian Standard Time (IST). It adds a Day Trader analyst whose single-session plans are stored as data and scored after the session by deterministic code on 5-minute bars, net of charges. And it adds data-integrity controls in the shared tool, vendor and memory layers that keep later data out of past-dated runs and report gaps as unavailable rather than leaving them silent.

We describe the system from its code and report the quality-assurance process. On 30 October 2026, eight multi-agent hunt, fix and review workflows reported 126 reproduced problems and recorded 115 fixes, and the passing test count grew from 1,412 upstream to 3,139. A flaky test traced to a leaked worker thread serves as a worked example. A case study covers nine paper runs on 30 October 2026 with no real orders. One intraday plan traded: TCS.NS was stopped out for a net loss of ₹253.63 (−1.48R). One plan for 2 November is pending, and the other runs stood aside or are swing ratings not yet scored. These runs are anecdotes, not evidence that the recommendations add value. Finally, we summarise a verified regulatory and licensing review. Under SEBI's Research Analyst Regulations the outputs are research services, and the data sources and two of the model providers are not licensed for a product; the review concludes that the product cannot be sold in India today. We outline a registered, human-reviewed path and a pre-registered forward evaluation with kill criteria.

Keywords: multi-agent systems; large language models; equity research; NSE; BSE; look-ahead bias; intraday trading; software verification; SEBI; research analyst regulation.

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1 Introduction

1.1 Motivation

Multi-agent LLM systems for finance imitate the division of labour in an investment firm. In TradingAgents [1], analysts gather market, news, sentiment and fundamental evidence; researchers argue the bull and bear cases; a trader proposes a position; a risk team debates it; and a portfolio manager decides. The upstream authors report strong results from a three-month backtest on US large-cap stocks [1]. The framework is open source under the Apache License 2.0 [43], which makes it a natural starting point for other markets.

India is not a change of ticker suffix. NSE and BSE run on IST, keep their own holiday lists and special sessions, and change their closing procedure twice in 2026: a closing auction from 3 August 2026 [29], [32], and a revised auction, already announced, from 7 December 2026 [30]. Prices are quoted in rupees with lakh and crore grouping. Transaction taxes and charges weigh heavily on intraday trading: SEBI's own study found that 71% of individual intraday traders in the equity cash segment made net losses in FY23, and that trading costs added 57% to the losers' losses [24]. Indian regulation also treats buy, sell and hold recommendations, price and stop-loss targets and intraday calls as research services that only a registered research analyst may provide [25].

Evaluation is a further problem. LLMs remember market history from their training data, and instructing them to ignore it does not work [6], [7]. Backtests inside a model's training window are therefore inflated [15], and longer, broader or live tests have removed much of the edge reported for LLM trading agents [16]–[19]. The only independent live test of the upstream design that we found shows it beating buy-and-hold in 5 of 20 single-model runs over two months, by our count from its table [19].

These facts shaped the engineering. The fork tries to make data honest at the point of use, to score plans by deterministic code rather than by the model's own account, and to treat only forward, time-stamped runs as informative. It was also built quickly with an AI assistant. That raises a software-quality question, which this report answers with a record of the checking process rather than with a claim of correctness.

1.2 Contributions

1. An adaptation of a multi-agent equity-research framework to NSE and BSE: symbol resolution, 2026 calendars and session rules applied by date, rupee conventions, configurable charge and capital-gains profiles, Indian benchmarks and IST news windows (Section 4).
2. Data-integrity and look-ahead controls placed in shared tool, vendor and memory layers rather than in prompts, which turn each data gap into an explicit "unavailable" signal (Section 5).
3. An intraday mode in which a plan is frozen as structured data at decision time and scored after the session by deterministic code on 5-minute bars, net of Indian charges, under assumptions that lean against the plan (Section 6).
4. A documented adversarial quality-assurance process run by teams of AI agents, with its statistics, its test-suite growth and a worked root-cause analysis (Section 7).
5. A case study of nine paper runs on one trading day, reported in full: a losing trade, the data defects the runs exposed and the fixes that followed (Section 8).
6. A summary of a verified review of Indian regulatory and licensing constraints, a recommended path to a lawful product, and a pre-registered evaluation protocol with kill criteria (Sections 9 and 11).

1.3 Scope and claims

This is a systems report and a case study. It does not claim that the system's ratings or plans make money, and the evidence in it could not support such a claim: the live runs cover one trading day, one executed paper trade, one pending trade plan and three swing ratings not yet scored. The report has not been peer reviewed. Nothing in it is investment advice or legal advice.

1.4 Provenance and conventions

Every number in this report comes from a file, a command output or the code, and was checked against its source on 30 October 2026. A figure that could not be confirmed is omitted or marked unknown. Where a value was recomputed independently, the text says so.

The code described is the main working tree of the fork as of 22:20:42 IST on 30 October 2026: the public head commit `27e6744` plus five rounds of uncommitted fixes, 95 files changed relative to that commit (68 modified and 27 new). Code and tests did not change after that time while this report was prepared. Documentation files were still being edited and are not used as evidence. File references such as `graph/setup.py` are relative to the package or repository root, and `yahoo/` and `vendors/` abbreviate `dataflows/vendors/yahoo/` and `dataflows/vendors/`. All times are IST unless marked otherwise. "The model" means the LLM.

The repository is public at https://github.com/abhishekjunior1/TradingAgents under Apache-2.0. GitHub does not record it as a fork of TauricResearch/TradingAgents, so we call it an independent fork. It derives from upstream release v0.7.1 (merge commit `c454491`, tagged `v0.7.1`), and its version string is still 0.7.1. Two commits have been pushed. Both were committed on 30 October 2026: `c432e28` at 13:16:35 and `27e6744` at 14:03:49. Between the upstream release and `27e6744`, 121 files changed, with 19,867 lines added and 244 removed.

2 Background and related work

2.1 The upstream framework

Xiao et al. [1] model an LLM "trading firm". An analyst team (fundamental, sentiment, news and technical) feeds bull and bear researchers who debate for a set number of rounds under a facilitator; a trader proposes a trade; a three-member risk team (risk-seeking, neutral and conservative) debates it; and a fund manager approves it. Agents pass structured reports and use natural-language dialogue only in debates. The authors motivate the design by two weaknesses of earlier agents: a "Lack of Realistic Organizational Modeling" and "Inefficient Communication Interfaces", which they liken to a "telephone effect". Backbone models are exchangeable; the paper used gpt-4o-mini and gpt-4o for quick thinking and o1-preview for deep thinking.

The evaluation is a backtest from 1 January to 29 March 2024. Its results table covers AAPL, GOOGL and AMZN, with cumulative returns of 26.62%, 24.36% and 23.21% and Sharpe ratios of 8.21, 6.39 and 5.60. The authors note that "the highest Sharpe Ratio exceeds our expected empirical range", attribute it to few pullbacks in the period, and explain that the test lasted three months because each prediction needed "11 LLM calls & 20+ tool calls". A reading of version 7 made for this report (ours, not the authors') found no mention of commissions, slippage or transaction costs, no repeated runs and no significance tests. The window spans about 61 US trading days by our count, and the abstract's claim of improved maximum drawdown is not borne out by the table, where TradingAgents has the lowest drawdown on none of the three stocks. In the study's favour, the documented knowledge cutoff of the models it used, October 2023 [46], precedes the test window; the paper itself does not make this point.

This report uses [1] for the architecture the fork inherits. It does not cite it as evidence that the fork's ratings add value: the upstream evaluation is short, cost-free and limited to US large caps, and it says nothing about Indian markets or intraday plans.

2.2 Other LLM agent frameworks for finance

FinMem [2] is a single trading agent with a layered memory and a configurable "character". FinAgent [3] is a multimodal agent that reads prices, text and charts and reflects at two levels. FinRobot [4] is an open-source platform built around a "Financial Chain-of-Thought" rather than a trading evaluation. FinCon [5] is the nearest design relative of TradingAgents: a manager-analyst hierarchy with a risk-control component that updates "investment beliefs" by verbal reinforcement. A later benchmark [16] lists their original evaluation windows: 3 months and 3 symbols for TradingAgents, 6 and 5 for FinMem, 6 and 6 for FinAgent, and 8 and 8 for FinCon.

2.3 Look-ahead bias and memorisation

LLMs recall exact economic and financial values from before their training cutoff. Lopez-Lira, Tang and Zhu report that "instructions to respect historical boundaries fail to prevent recall-level accuracy", and that masking fails because models reconstruct entities and dates from minimal context [6]. Sarkar and Vafa find look-ahead bias in two forecasting applications and discuss the limits of prompting-based remedies [7]. Gao, Jiang and Yan estimate a "lookahead propensity" for each firm and date that is positive in-sample and collapses to near zero after the cutoff [8]. Ludwig, Mullainathan and Rambachan state the condition formally: valid predictive conclusions require "no training leakage" between the model's training data and the researcher's sample [9]. DatedGPT trains models with annual cutoffs and measures a "lookahead premium" of 26.4 basis points per standard deviation [10]. Levy reports significant look-ahead bias, and poor numerical reasoning, in commercial models [13].

The evidence is not uniform. Glasserman and Lin find that, in-sample, distraction by company names mattered more than look-ahead for headline sentiment [12], and He et al. find the bias modest in a news-return test with chronologically consistent models [14]. A position paper surveying 164 studies found that none of five biases (look-ahead, survivorship, narrative, objective and cost) was discussed in more than 28% of them [11].

For this report the consequence is direct. Restricting the data a tool returns to the analysis date, which the fork does (Section 5), removes one channel of leakage but cannot remove what the model learned in pre-training. Past-dated runs of the fork are therefore diagnostic only.

2.4 Out-of-sample and live evaluations

Li et al. call the pattern a "profit mirage": back-tested returns "evaporate once the model's knowledge window ends" [15]. FINSABER backtests LLM strategies over two decades and more than 100 symbols and finds that reported advantages "deteriorate significantly" under a broader and longer evaluation, with strategies too conservative in bull markets and too aggressive in bear markets [16]. StockBench trades 20 large US stocks over 82 trading days in 2025, without modelling costs, and finds that "most models struggle to outperform the simple buy-and-hold baseline" [17]. LiveTradeBench runs 50-day live evaluations of 21 models and finds that high general-benchmark scores "do not imply superior trading outcomes" [18].

Agent Market Arena [19] is the only independent live test of the upstream design that we found. It ran TradingAgents (as "TradeAgent") live in August and September 2025 on TSLA, BMRN, BTC and ETH with five backbone models. On TSLA its cumulative return ranged from −38.72% to 21.91% depending on the model, against 46.88% for buy-and-hold. By our count from the paper's table, it beat buy-and-hold in 5 of 20 single-backbone runs. The paper's daily-return formula has no cost term.

Positive forward evidence exists for narrower tasks. GPT-4 scores of post-cutoff headlines capture the initial, non-tradable market reaction with about 90% portfolio-day hit rates, although strategy returns decline as LLM adoption rises [20]. Chen and Pu collected daily ratings of Russell 1000 stocks "at the current edge of time" from April 2025 and report alpha only for the top-ranked stocks [21]. Pelster and Val ran a live experiment with ChatGPT-4 and report a positive correlation between its ratings and later earnings announcements [22]. In these designs every prediction is recorded before its outcome exists. They are the model for the evaluation we propose in Section 11.

2.5 Indian market context

The only evaluation of LLMs on Indian equities that we found is Sodha's walk-forward study on NIFTY constituents at a one-day horizon. The ranking quality of a "thinking" model deteriorated as the universe grew, and results under transaction costs "do not support a net advantage" [23]. SEBI's study of intraday trading by individuals gives the base rate quoted above: 71% of individual intraday traders in the equity cash segment made net losses in FY23, rising to 80% among very frequent traders. Trading costs added 57% to the losses of loss-makers and consumed 19% of the profits of profit-makers [24].

On regulation, SEBI's Research Analyst (RA) Regulations define "research services" to include buy, sell and hold recommendations, price and stop-loss targets, model portfolios and trading calls, and define trading calls to include intraday and non-personalised recommendations [25]. A research analyst using AI tools is solely responsible for research based on their output and must disclose the extent of AI use [25], [26]. Advertisements may refer to past performance only if a Past Risk and Return Verification Agency (PaRRVA) has verified it [26], a role SEBI gave CARE Ratings with NSE as data centre [33]. NSE's regular equity session runs from 09:15 to 15:30 IST [31].

2.6 Positioning

The literature points one way. Short backtests of LLM trading agents on a few US stocks are common; inside the training window they cannot be separated from memorisation; and longer or live tests shrink or remove the edge. This report therefore presents engineering for a new market and a case study, and proposes a forward evaluation, rather than reporting returns.

3 System overview

3.1 Architecture

The fork keeps upstream's single-process design, built on LangGraph, a library that expresses agent workflows as state graphs. Figure 1 shows its components. Runs start from a command-line interface (CLI), a Python API or a Claude Code skill. A graph orchestrator runs the agents. Tool-using agents call a fixed set of tools, every tool reaches outside data through one vendor router, and pure market modules supply calendars, position sizes and charges. Decisions are written to reports, a JSON state log and a Markdown memory log that later runs settle and learn from.

Configuration is a default dictionary built at import, with 36 settings that `TRADINGAGENTS_*` environment variables can override. Values are coerced to the default's type, and an invalid value stops the program at start-up. Each run's configuration is held in a context variable, so concurrent graphs in one process read their own vendor settings. The fork supports 22 LLM provider keys (5 native and 17 OpenAI-compatible), and its two model tiers, "quick" and "deep", may use different providers. One provider, `claude_code`, drives the local Claude Code command-line tool (`claude -p`) with no tools enabled and a credential-free child environment. All the live runs in Section 8 used it.

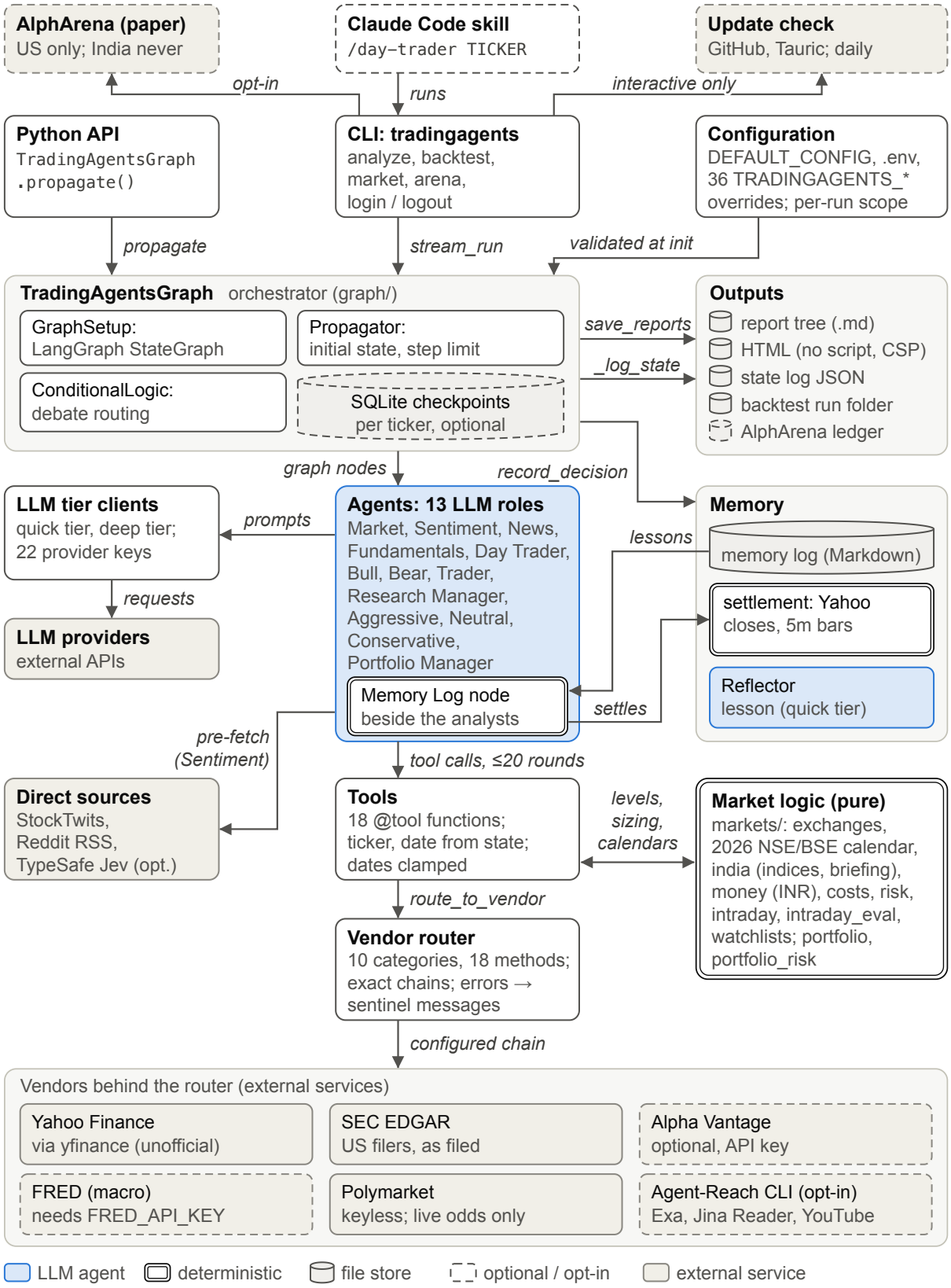


Figure 1. System architecture of the fork, drawn from the code. The legend distinguishes LLM agents, deterministic code, file stores, optional or opt-in components and external services. The India support and the look-ahead guards sit in the shared tool, router, market and memory layers, not in agent prompts. Sources: `cli/main.py`, `graph/trading_graph.py`, `graph/setup.py`, `agents/tools.py`, `dataflows/router.py`, `markets/`, `memory/`.

3.2 The agent pipeline

Figure 2 shows the pipeline for one decision. Every selected analyst and a Memory Log node start together from the graph's entry point; the code's comment states: "The analysts work at the same time". Each analyst is its own subgraph, with a private message history and an output schema that holds only its report, so parallel analysts never write the same state key. An analyst alternates between the model and its tools. After 20 tool rounds (the default cap), a final turn without tools asks it to write its report.

When all reports are filed, the Bull and Bear researchers debate, the Bull opening, for 2 × `max_debate_rounds` turns: two at the default depth of one. A Research Manager on the deep tier writes a plan with a five-tier recommendation (Buy, Overweight, Hold, Underweight or Sell). The Trader proposes a position. Three risk debaters (Aggressive, Conservative and Neutral) speak in turn for 3 × `max_risk_discuss_rounds` turns, and a Portfolio Manager on the deep tier makes the final decision. Only the Portfolio Manager reads lessons from past decisions. Table 1 lists the agents.

Outputs are structured: a research plan, a trader proposal and a portfolio decision, the last two with intraday variants. A failed structured call falls back once to free text, and a rating that cannot be read becomes the sentinel `REVIEW` rather than a default Hold.

Table 1. Agents, model tiers and tools. Thirteen roles run in the graph; the Reflector runs only when a decision is settled. Sources: `agents/`, `graph/setup.py`, `memory/reflection.py`.

Agent	Tier	Tools or inputs
Market Analyst	quick	`get_stock_data`, `get_indicators`, `get_verified_market_snapshot`, `get_market_context`
Sentiment Analyst	quick	No tools: pre-fetched news (7 days), StockTwits (30 messages) and Reddit RSS, optionally screened by an external service
News Analyst	quick	`get_news`, `get_global_news`, `get_macro_indicators`, `get_prediction_markets`; three web and video tools when Agent-Reach is switched on
Fundamentals Analyst	quick	`get_fundamentals`, `get_balance_sheet`, `get_cashflow`, `get_income_statement`, `get_insider_transactions`, `get_corporate_actions`
Day Trader (new in the fork)	quick	`get_intraday_snapshot`, `get_verified_market_snapshot`, `get_market_context`
Bull and Bear Researchers	quick	All analyst reports; no tools
Research Manager	deep	The debate; no tools
Trader	quick	Research plan and market report (swing) or Day Trader report and session brief (intraday)
Aggressive, Conservative and Neutral debaters	quick	Analyst reports, the trader's proposal and any portfolio
Portfolio Manager	deep	All of the above, plus lessons from the memory log
Reflector (settlement only)	quick	A settled outcome; writes a two-to-four-sentence lesson

3.3 Tools, router and vendors

The fork offers agents 18 tools. Each dated tool takes the ticker and the analysis date from the graph state, not from the model, so the model can choose neither. Tools reach data through a router with 10 categories, 18 methods and 32 method-vendor implementations over six routed vendors: Yahoo Finance (through the unofficial `yfinance` library), SEC EDGAR, Alpha Vantage, FRED, Polymarket and the opt-in Agent-Reach command-line tool (Exa web search, Jina Reader and YouTube).

The configured vendor chain is the chain: there is no silent fallback to a vendor the user did not choose. Vendor failures are typed (no data, unavailable, not configured) and reach the agents as sentinel strings such as `DATA_UNAVAILABLE` and `NO_DATA_AVAILABLE`, with news gaps worded as "not an absence of news". Optional categories degrade to a sentinel; a hard failure in a core category raises an error. Some sources

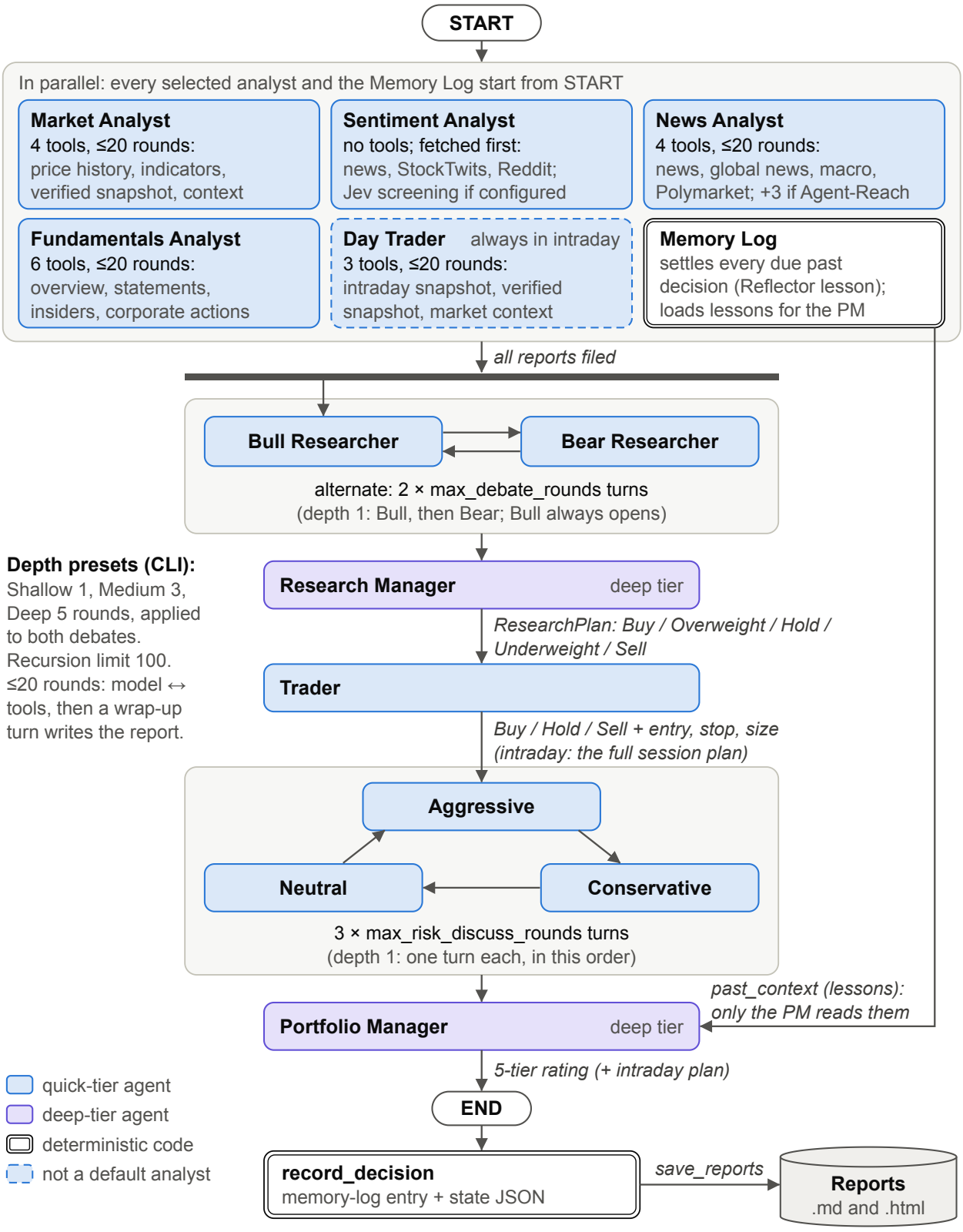


Figure 2. The agent pipeline for one decision. The analysts and the Memory Log node run in parallel; each analyst loops between the model and its tools for at most 20 rounds. The Bull and Bear researchers alternate for 2 × `max_debate_rounds` turns, and the risk debaters speak for 3 × `max_risk_discuss_rounds` turns. Only the Portfolio Manager reads past lessons. The CLI's depth presets (Shallow 1, Medium 3, Deep 5) set both debate lengths. Sources: `graph/setup.py`, `graph/conditional_logic.py`, `cli/prompts.py`.

sit outside the router: StockTwits and Reddit for the Sentiment Analyst, an optional screening service for social posts, and direct Yahoo calls for instrument identity, the verified snapshot, session planning and settlement prices.

3.4 Entry points, outputs and memory

The CLI (Figure 3) runs one analysis, with flags that answer its interactive questions, including `--exchange nse|bse`, `--horizon swing|intraday`, `--benchmark` and `--vendor`. Its other commands are `backtest`, which runs a grid of tickers and dates into a memory log of its own; `market`, which prints exchange status and market context without calling a model; `arena`, for an opt-in paper-trading service that never receives Indian listings; and `login` and `logout` for one provider. Run without a terminal, the CLI lists every unanswered question before any model call.

Each run writes three kinds of output. The first is a report tree: one Markdown file per section, plus `complete_report.md` and a self-contained HTML page with no scripts and a content-security policy that blocks every fetch. The second is a JSON state log whose recorded settings exclude endpoints, keys and paths. The third is an entry in an append-only Markdown memory log. The log keeps one entry per ticker, date and horizon. Writes hold a file lock, rewrites go through a temporary file and an atomic replace, and model text is escaped so that it cannot forge the log's structure lines.

A later run's Memory Log node settles due decisions for every ticker. A swing rating is measured over five trading days from the first close after the analysis date, against the benchmark stored with the decision. An intraday plan is replayed on its session's 5-minute bars (Section 6). A Reflector writes a short lesson, and up to five same-ticker and three cross-ticker lessons reach later Portfolio Managers. The `backtest` command is swing-only. It gives each sweep its own memory log, switches off the paper-trading service and the web-research vendor, and prints a summary that calls its own figures "indicative rather than repeatable".

3.5 What changed relative to upstream

Table 2 compares the upstream release with the fork's working tree, measured with the same counting script. The code grew by about three-quarters (76% more physical lines and 80% more code lines), and the passing test count more than doubled.

Table 2. Size of upstream release v0.7.1 and of the fork's main working tree (code as of 22:20:42 IST, 30 October 2026). Code lines exclude comment-only and docstring lines. Pass counts are from re-runs in a clean environment; in both, one test is skipped because it needs an optional package and one integration test is deselected by default.

Measure	Upstream v0.7.1 (`c454491`)	Fork working tree
LLM agent roles in the graph	12	13
Tools offered to agents	12	18
Router categories / methods / routed vendors	6 / 11 / 5	10 / 18 / 6
Settings overridable by `TRADINGAGENTS_*` variables	20	36
Python files in `tradingagents/` and `cli/`	93	122
Physical lines / code lines	13,571 / 8,785	23,899 / 15,773
Test files / test functions	88 / 1,009	134 / 1,905
Collected test items	1,414	3,141
Tests passed	1,412	3,139



Figure 3. Top-level help of the fork's CLI, captured at 22:14:31 IST on 30 October 2026 in a clean environment and rendered at 110 columns. It lists the options of the analysis command, including `--exchange`, `--horizon`, `--benchmark`, `--vendor` and `--arena`, and the commands `backtest`, `market`, `login`, `logout` and `arena`. The app was invoked directly under the program name `tradingagents` because the console script was not on the clean `PATH`; the output is the same as `tradingagents --help`.

3.6 Cost of a run

The number of model calls follows from the code; it was not measured. For a default swing run with four analysts and debate depth one, the count is $t + 12 + F + S$. Here t is the analysts' tool-calling turns, at most 20 for each of the three tool-using analysts and so at most 60; F is at most four free-text retries of failed structured calls; and S is one reflection for each past decision the Memory Log node settles. The range is therefore $12 + S$ to $76 + S$ calls. After the analysts, eight calls run strictly in sequence, two of them on the deep tier. An intraday run through the `/day-trader` skill costs $t + 11 + F + S$. For compari-

son, the upstream paper reports 11 LLM calls and more than 20 tool calls per prediction [1]. Three intraday runs on 30 October took 9 min 43 s, 10 min 15 s and 7 min 52 s of wall-clock time. These are single observations, not a benchmark.

4 Adapting to Indian markets

4.1 Symbols and exchanges

Two exchange records carry the Indian specifics. NSE uses the Yahoo suffix `.NS` and the benchmark NIFTY 50; BSE uses `.BO` and SENSEX. Both use Asia/Kolkata time and rupees, with a pre-open from 09:00 and a regular session from 09:15 to 15:30 [31]; the code treats the day's close as final at 16:00. Only these two exchanges get session and holiday checks; other listings are handled as upstream handled them.

A normaliser maps broker notations to Yahoo's. `NSE:RELIANCE`, `RELIANCE.NSE`, `RELIANCE-EQ` and `NSE:RELIANCE-EQ` all become `RELIANCE.NS`; `BSE:` and `BOM:` prefixes become `.BO`; and index names such as `NIFTY` and `BANKNIFTY` become Yahoo index symbols. A bare symbol takes the default exchange's suffix, set by `--exchange` or an environment variable. Six-digit BSE scrip codes are refused because Yahoo does not serve them. Each listing has a "day" in its own zone: an NSE or BSE listing's date is IST's, and a plain US listing's date is New York's.

4.2 Calendars and dated session rules

The 2026 calendar lists 16 weekday holidays for NSE, taken from two NSE circulars cited in the code; BSE's list is identical. It also lists two special sessions: the Union Budget session on Sunday 1 February 2026, and Muhurat trading on Sunday 8 November 2026 (pre-open 14:05, trading 14:20–15:30). A year that is not listed is not guessed: only weekends count as closed, and the status line says so. Settlement holidays are not modelled.

Session rules change by date, as SEBI's circulars require (Table 3), and a run receives the rules in force on its own date. The session model distinguishes closed, pre-open, open and closing states. It treats a day as complete only at 16:00 IST, so a run during the session is told that today's bar is still forming. RBI Monetary Policy Committee dates for 2026–27 are given to a run only if it is dated on or after the schedule's publication on 23 March 2026, so a past-dated run does not learn a schedule that did not yet exist.

Table 3. Equity cash-market session rules the fork applies by date, with their sources. Pre-open is 09:00–09:15 throughout [29], and the close is final at 16:00.

Period	Stocks	Continuous trading	Close	Source
Before 3 Aug 2026	All	09:15–15:30	Closing price is the volume-weighted average of the last 30 minutes	[29], [31]
3 Aug – 6 Dec 2026	With derivatives	09:15–15:15	Closing auction 15:15–15:35, within 3% of the 15:00–15:15 average; post-close 15:50–16:00	[29], [32]
3 Aug – 6 Dec 2026	Others	09:15–15:30	15:00–15:30 average	Code (Figure 8, notes)
From 7 Dec 2026	With derivatives	09:15–15:30	Closing auction 15:30–15:40; post-close 15:50–16:00	[30]
Special sessions, 2026	All	Budget, Sun 1 Feb: 09:15–15:30 (pre-open 09:00); Muhurat, Sun 8 Nov: 14:20–15:30 (pre-open 14:05)	–	Code (`markets/india_calendar.py`)

4.3 Money, charges and tax

Rupee amounts use Indian digit grouping (12345678 becomes 1,23,45,678), with lakh and crore in compact form, and the Fundamentals Analyst is told to keep figures in rupees and give large ones in crore. Charges are configurable profiles (Table 4) dated 29 October 2026. Every field can be overridden, and a mistyped profile name is refused at start-up. Swing (delivery) decisions use `india_equity_delivery` by default; intraday plans are sized and scored with `india_equity_intraday`. Listings outside India carry no modelled intraday charges.

A capital-gains rule set for listed Indian equity treats holdings of more than 12 calendar months as long-term. Short-term gains are taxed at 20%, and long-term gains at 12.5% above ₹1.25 lakh a year, with surcharge and cess excluded.

Table 4. Built-in Indian charge profiles (`markets/costs.py`, dated 29 October 2026). The rates are examples to be checked against a broker; a third profile, `none`, applies no charges.

Charge	`india_equity_delivery`	`india_equity_intraday`
Brokerage	None	0.03% of order value, capped at ₹20 per order
Securities transaction tax (STT)	0.1% on buy and on sell	0.025% on sell only
Exchange transaction charge	NSE 0.00307%, BSE 0.00375%	Same
SEBI fee	₹10 per crore	Same
Stamp duty (buyer)	0.015%	0.003%
GST	18% of brokerage, exchange and SEBI charges	Same
Depository charge per sell order	₹15.34	None

These charges matter. In the TCS.NS paper trade of Section 8, round-trip charges of ₹82.03 added 48% to a gross loss of ₹171.60.

4.4 Benchmarks, indices and market context

The code knows 25 Indian indices with their Yahoo symbols and aliases. Six are marked as having no usable Yahoo history, and nothing computes a return from them. Each decision's default benchmark comes from a suffix map with 21 entries (`.NS` to NIFTY 50, `.BO` to SENSEX, and so on), and a run may choose its own with `--benchmark`. A benchmark check refuses 36 volatility-index symbols, including India VIX, and indices without Yahoo history. The sector index is the NIFTY sectoral index closest to Yahoo's present-day sector and industry, an approximation that the output states.

For an NSE or BSE run, the market-context tool gives the NIFTY 50, SENSEX, NIFTY Bank, India VIX, the closest sector index, USD/INR and Brent crude. Each row has its own last close date and changes over 1, 5, 21 and 63 sessions, and an Indian series whose expected close is not yet published is labelled late. Relative performance is computed over shared sessions. Every series comes through the same price loader as the agents' prices, so no close after the run date can appear. Figure 4 shows the same table printed by the `market` command, with a watchlist of the five stocks in the case study. The software's own source note reads: "Yahoo lists NSE quotes as real-time and BSE quotes as 15-minute delayed, and says its data is not intended for trading".

4.5 News windows and India news

News, social posts and web results are filtered to a window [start, end + 1 day) in the listing's own zone. A comment in the code gives the reason: "a past NSE run dated the 29th must not read what India published after its own midnight, 18:30 UTC". Undated items are kept only when the window reaches the present.

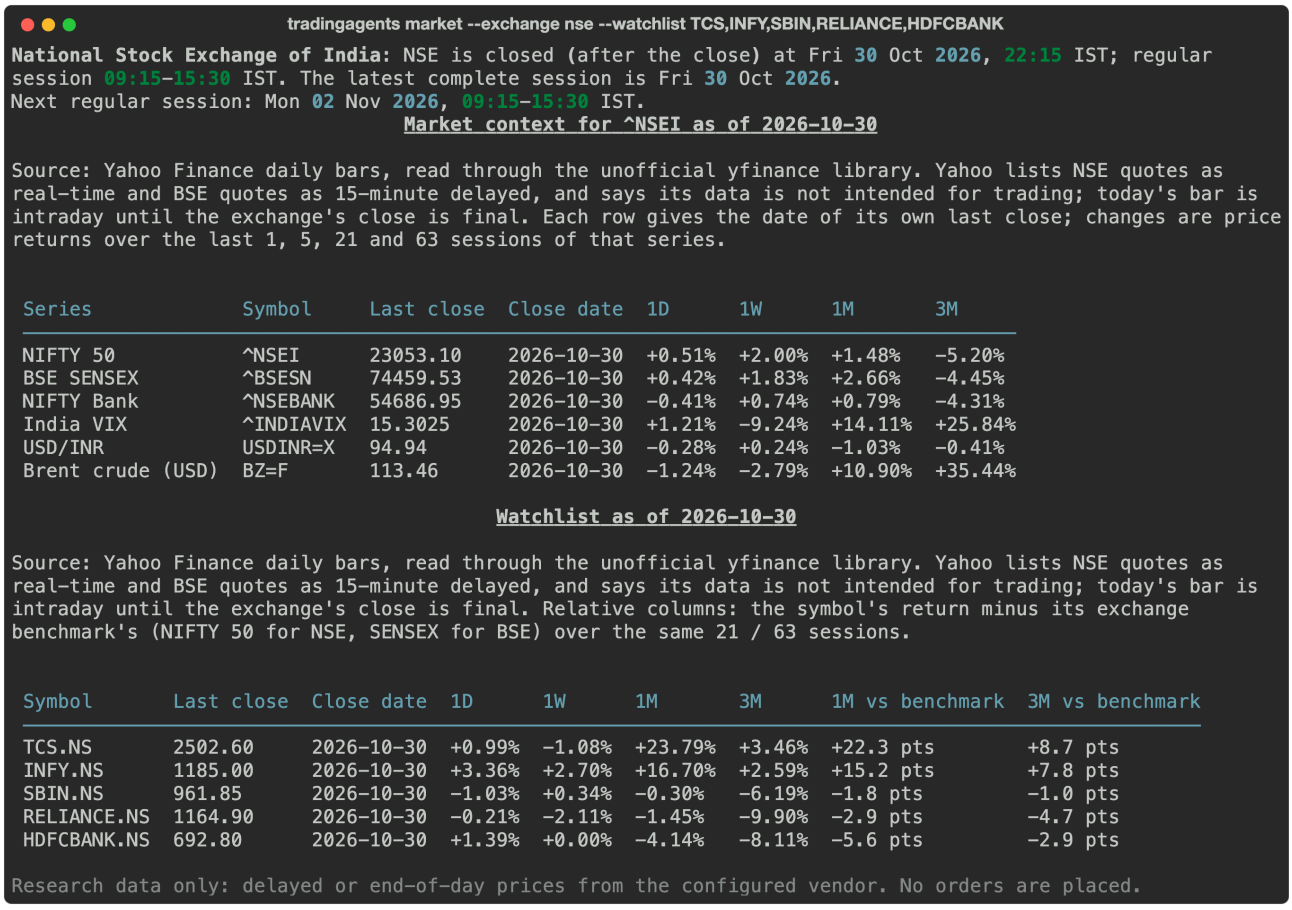


Figure 4. `tradingagents market --exchange nse --watchlist TCS,INFY,SBIN,RELIANCE,HDFCBANK`, captured at 22:15:05 IST on 30 October 2026, after the close. The session line reports NSE closed, with the next session on Monday 2 November. The market-context table lists NIFTY 50 (23053.10), SENSEX, NIFTY Bank, India VIX, USD/INR and Brent with their own close dates. The watchlist gives each stock's last close (TCS.NS 2502.60, INFY.NS 1185.00, SBIN.NS 961.85, RELIANCE.NS 1164.90, HDFCBANK.NS 692.80) and its return relative to NIFTY 50. No model is called. Data: Yahoo Finance through `yfinance`.

For an NSE or BSE instrument, the global-news tool reports two groups, India and Global. India items are searched with the terms rupee, RBI and India. They are kept only if a strong term (such as India, SEBI, Sensex or Dalal Street) or two weak terms (such as rupee, RBI, Nifty, NSE or BSE) appear as whole words, or if Yahoo tags an Indian listing, an Indian index or the rupee. An empty result is reported as unavailable coverage, not as an absence of news.

4.6 Prompts, disclaimers and what the system cannot see

A market briefing gives agents rules without live numbers: IST, the sessions in force on the run date, T+1 settlement, price bands and market-wide circuit breakers, results deadlines, the MPC and the Budget. It also lists what the system cannot see: foreign and domestic institutional flows, promoter holdings and pledges, futures and options open interest, and exchange announcements. Evidence rules tell every agent to tie each number to a tool and a date.

Indian companies report cash flow half-yearly under SEBI's listing rules, so a request for quarterly cash flow is answered with that explanation rather than with an empty result that could read as an invalid symbol. Reports on NSE and BSE listings state that they are not research or advice from a SEBI-registered research analyst or investment adviser. The opt-in paper-trading service, AlphArena, never receives Indian listings.

5 Data integrity and look-ahead controls

5.1 Design principle

The fork's integrity controls follow one principle. Enforce in code what can be enforced, at a layer every agent shares, and tell the model explicitly when data is missing rather than leaving a silence for it to fill. Figure 5 shows the layers, and Table 5 lists the guards.

5.2 The guards

Table 5. Data-integrity and look-ahead guards in the fork's code (paths relative to `tradingagents/`; `yahoo/` stands for `dataflows/vendors/yahoo/`).

Guard	What it does	Code
Ticker and date pinned	Tools read the ticker and analysis date from graph state; the model cannot choose them	`agents/tools.py`
Date clamps	Any requested date is capped at the run date; a later window is moved back	`dataflows/date_window.py`
Price cut-off	Daily rows after the run date are dropped before use; the verified snapshot re-applies the cut-off	`yahoo/ohlcv.py`, `yahoo/snapshot.py`
Stale data refused	A latest row more than 10 days before the requested date raises a no-data error	`yahoo/ohlcv.py`
Same-day cache	A cache written today serves a current-day request for 900 seconds only, because Yahoo's daily candle is partial	`yahoo/ohlcv.py`
Forming-bar labels	A bar of a session still trading is labelled "its Close is the latest price, not a close", and its indicators "provisional"	`yahoo/market.py`, `yahoo/snapshot.py`
Short history	An indicator with too few sessions (200 for the 200-day average, 15 for RSI and ATR) returns "N/A: not enough price history" instead of a default value	`yahoo/market.py`, `markets/risk.py`
Holiday filler rows	A flat, zero-volume Yahoo row on a calendar holiday is dropped, and the output says so	`yahoo/ohlcv.py`
Missing trading days	NSE or BSE trading days with no Yahoo row are listed as missing, "not holidays"	`yahoo/market.py`
Adjusted and unadjusted prices	Daily prices are labelled as adjusted for splits and dividends; intraday bars are unadjusted; levels taken from daily rows are labelled	`yahoo/market.py`, `yahoo/intraday.py`
Intraday bars only once ended	Cut-off is the earliest of now, the fetch time minus the feed delay (15 minutes for BSE) and the session close of the run date	`yahoo/intraday.py`
News windows	Windows in the listing's zone; undated items only for current runs	`dataflows/date_window.py`
Live-only data withheld for past dates	Company profile, statements dated only by period end, insider trades dated by execution, prediction-market odds, upcoming corporate events and live web pages	`dataflows/date_window.py`, vendors
Identity	Past-dated runs get the current company name, labelled as such, without sector or industry	`agents/context.py`
Memory point in time	A past-dated run reads only lessons resolved before its date, or on that date when the markets involved close with the run's own market	`memory/log.py`, `graph/trading_graph.py`
Per-decision benchmark	Settlement uses the benchmark that the decision's own run used	`memory/settlement.py`
Unreadable decisions	`REVIEW`, never Hold; backtests count it as unscored	`agents/rating.py`, `backtest.py`

5.3 The verified market snapshot

The verified market snapshot is a deterministic tool built from Yahoo daily bars. It returns the latest row, 11 indicators (10-day EMA, 50- and 200-day SMA, RSI, the Bollinger middle, upper and lower bands, MACD with its signal and histogram, and ATR) and up to 30 recent closes. It reads values as reported,

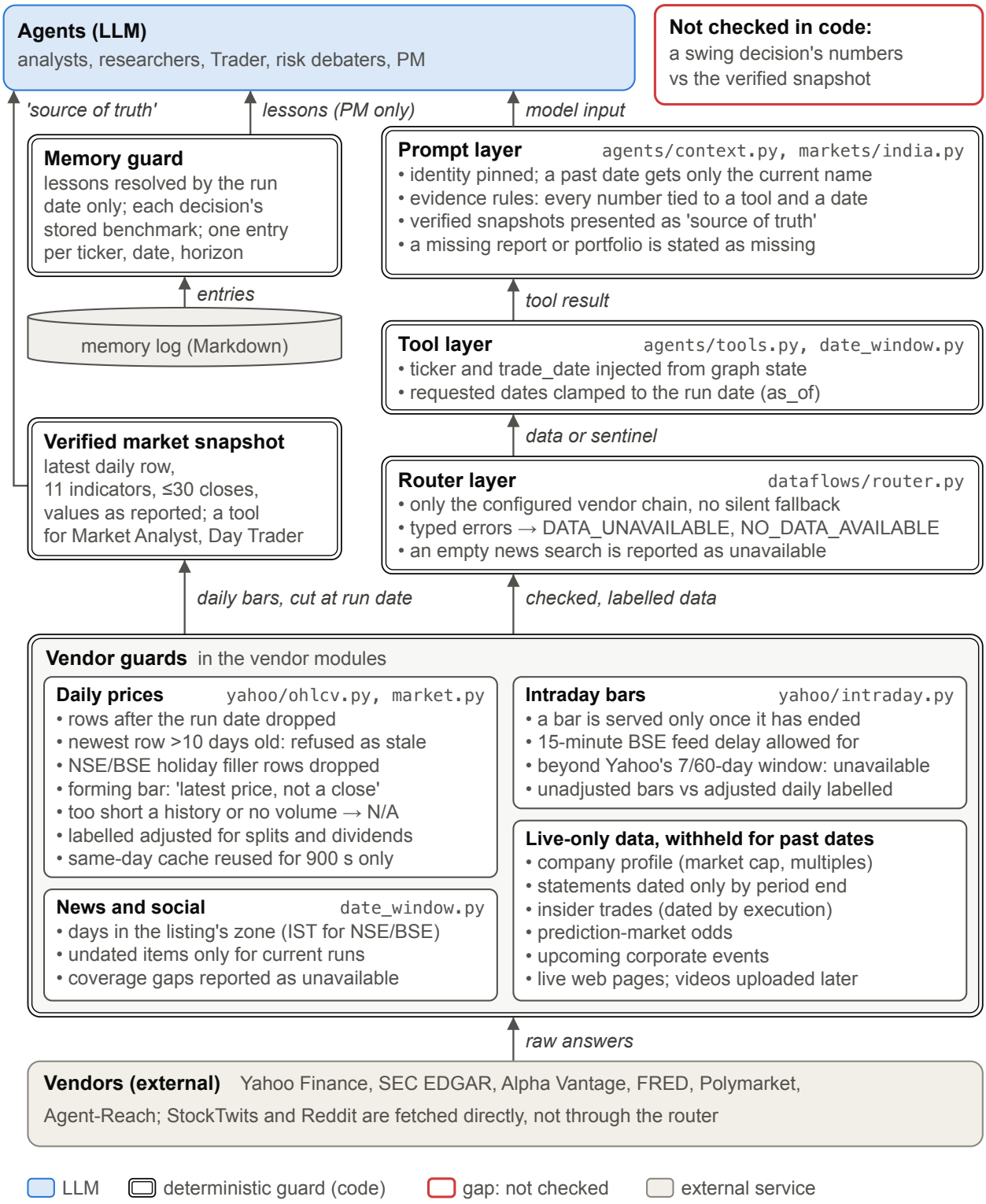


Figure 5. Data-integrity guards between the vendors and the agents: a prompt layer; a tool layer that pins the ticker and clamps dates; a router that turns failures into explicit sentinels; vendor guards for daily prices, intraday bars, news and live-only data; and a memory guard. The verified snapshot is an instruction to the model, not a check on its output, and nothing in code compares the figures in a swing decision with the verified data. Sources: `agents/tools.py`, `dataflows/date_window.py`, `dataflows/router.py`, `dataflows/vendors/`, `memory/log.py`.

without filling gaps, and ends with an instruction to treat it as the source of truth (Figure 6). For an NSE or BSE session still trading, its headings say the values are not final.

get_verified_market_snapshot: SBIN.NS, look_back_days=10

```
## Verified market data snapshot for SBIN.NS

- Requested analysis date: 2026-10-30
- Latest trading row used: 2026-10-30
- Daily rows behind the indicators: 1235, from 2021-11-01
- Rows after the requested analysis date are excluded before verification.
- Prices: adjusted by Yahoo for splits and dividends: rows dated before an ex-dividend date, even one after
the dates asked for, are scaled down by that dividend, so they can sit below the exchange's printed prices and
unadjusted figures such as the intraday bars or a quote's 52-week range.

### Latest verified OHLCV row

| Field | Value |
|---|---:|
| Open | 971.85 |
| High | 971.85 |
| Low | 959.60 |
| Close | 961.85 |
| Volume | 7507038 |

### Verified technical indicators (latest row)

| Indicator | Value |
|---|---:|
| close_10_ema | 961.109 |
| close_50_sma | 989.604 |
| close_200_sma | 1031.00 |
| rsi | 47.3529 |
| boll | 958.06 |
| boll_ub | 975.719 |
| boll_lb | 940.401 |
| macd | -6.66673 |
| macds | -9.53583 |
| macdh | 2.8691 |
| atr | 17.68 |

### Recent verified closes (last 10 rows)

| Date | Close |
|---|---:|
| 2026-10-16 | 962.60 |
| 2026-10-19 | 973.70 |
| 2026-10-21 | 963.00 |
| 2026-10-22 | 970.00 |
| 2026-10-23 | 958.60 |
| 2026-10-26 | 960.80 |
| 2026-10-27 | 945.10 |
| 2026-10-28 | 956.80 |
| 2026-10-29 | 971.90 |
| 2026-10-30 | 961.85 |

Use this snapshot as the source of truth for exact OHLCV, price-level, and indicator-value claims. If another
tool output conflicts with it, flag the discrepancy rather than inventing a reconciled number. Do not claim
historical validation, support/resistance bounces, or exact percentage moves unless directly supported by tool
output with concrete dates and prices.
```

Figure 6. The verified market snapshot for SBIN.NS dated 30 October 2026, with a 10-day look-back, exactly as an agent receives it. Captured at 22:18:37 IST, after the close, from 1,235 adjusted daily rows starting 1 November 2021: close 961.85, 50-day SMA 989.604, 200-day SMA 1031.00, RSI 47.3529, ATR 17.68. These are final-close values; the 14:43 swing report in Section 8 quoted in-session values. Figure 18 in Appendix B shows the 30-row version.

5.4 Independent recompute of the snapshots

To check the intraday snapshot's arithmetic, we recomputed six snapshots taken during the 30 October session, for five NSE stocks and RELIANCE.BO. The recompute used the cached 5-minute and daily bars, pure pandas and none of the fork's code. All 95 compared figures matched: open, high, low, last, VWAP, volume, gap, relative volume, opening ranges, pivots P and R1, and the 5-minute ATR. The daily ATR(14) matched for the five NSE stocks, each computed by Wilder smoothing over 1,235 adjusted daily rows (Table 6). RELIANCE.BO had no daily row before 30 October, and its snapshot showed n/a.

The fork has two ATR implementations: the stockstats library in the verified snapshot and Wilder smoothing in the intraday snapshot. We did not compare them numerically.

Table 6. Daily ATR(14) in the intraday snapshots taken during the 30 October session, each matched by an independent recompute (Wilder smoothing, an exponential mean with α = 1/14, over the full adjusted history).

Listing	Daily ATR(14)	Daily rows
TCS.NS	73.5931	1,235
INFY.NS	42.185	1,235
HDFCBANK.NS	14.2621	1,235
RELIANCE.NS	27.2592	1,235
SBIN.NS	18.0939	1,235
RELIANCE.BO	n/a	No row before 30 Oct

5.5 A live data gap: RELIANCE.BO

At 14:09 on 30 October an intraday run on RELIANCE.BO found that Yahoo held a single daily bar for the listing. The Market Analyst reported:

> `get_stock_data` for 2026-08-01 → 2026-10-30 returned 1 record out of 61 expected trading days; the tool explicitly flagged: "no row for 61 of the 61 BSE trading days in this range... they are missing from Yahoo's data for this listing, not holidays."

The daily-data guard had been added earlier that day (commit `27e6744`, 14:03:49), prompted by the same one-bar listing, and all eight indicator calls in the run returned N/A. The verified snapshot, however, still printed indicator values computed from one bar. The analyst noticed:

> `rsi` defaulting to 50.00, and `macd`/`macds`/`macdh` defaulting to 0.00 — these are placeholder/degenerate values produced because only one data point exists … `atr` shows 19.90 … (High–Low = 1172.40–1152.50 = 19.90), but it is not a smoothed 14-day ATR and should not be used for stop-sizing.

The run stood aside. The snapshot was then given the same minimum-history rule as the indicator tool, in the first round of uncommitted fixes, and its test cites this run. The episode shows both sides of the design. An explicit gap message let the model reason correctly in this instance, while a tool that returned a plausible default number put a misleading figure in front of it.

5.6 Vendor revisions after the session

Yahoo revises intraday data after the session. The 09:15 bar for TCS.NS on 30 October showed a volume of 135,481 when fetched at 14:26 and 0 when fetched at 22:31. Of the 358 bars common to both fetches over 26–30 October, only that bar and the 14:25 bar, still forming at 14:26, differed. Recomputing the 13:30 snapshot from the later fetch gives a VWAP of 2512.92, a volume of 1,730,842 and a relative volume of 0.79. The bars served in the afternoon give 2512.71, 1,866,323 and 0.85, which match the snapshot exactly. A replay of a plan is therefore only as reproducible as the data captured with it, which is one reason the assessment in Section 9 recommends storing raw tool outputs with each run.

5.7 What is not checked

Two gaps remain. First, no code compares the numbers in a swing decision's text with the verified snapshot: the snapshot is an instruction, not a validator. The only machine checks on a decision's figures are the intraday plan checks (Section 6.3), which add notes and never correct the plan. Second, the guards restrict what tools return; they cannot remove what the model learned before its training cutoff [6], [7]. A search of the code found no check of the model's training cutoff against the run date.

6 Intraday planning and settlement

6.1 Life of a plan

An intraday run differs from a swing run in three ways. It always includes the Day Trader analyst; its Trader and Portfolio Manager write a plan for a single session; and that plan is scored after the session by code rather than by a model. Figure 7 shows the life of a plan.

6.2 Session context and the snapshot

At the start of a run, the code fixes the session the plan is for. A past date plans the next trading day. Today plans the rest of today's session while it is open or yet to open, and otherwise the next trading day. The code also fixes the exit-by time and a reference price. The exit-by time is the configured square-off, by default 15:10 IST for NSE and BSE, which matches the square-off times of some Indian brokers noted in the configuration. The reference price is the close of the newest complete 5-minute bar. These values reach the Day Trader, Trader and Portfolio Manager in a session brief that also explains when an entry is a stop order and when it is a limit order.

The Day Trader's main tool is the intraday snapshot (Figure 8). From bars that had ended by the cut-off, it computes:

- the last session's prices and its volume relative to earlier sessions;
- for the plan's session so far, the open, high, low, last price, VWAP, gap, official gap, relative volume to the same time of day and the 15- and 30-minute opening ranges;
- classic floor pivots from the last session;
- ATR(14) on the intraday bars and on the daily rows before the session;
- paper sizing examples, long and short, with stops one and two ATRs away.

Sizing gives the largest number of shares whose stop-out loses no more than the configured risk percentage of capital, including round-trip charges, capped by capital times the leverage cap divided by the entry price. The defaults are ₹1,00,000 (one lakh) of paper capital, 0.5% risk and no leverage.

6.3 The plan and its checks

The Portfolio Manager's intraday decision has labelled fields: Direction (long, short or none), Entry Price, Stop Loss, Target Price, Exit By, Position Size (shares) and Invalidation. A free-text fallback is parsed by the same labels and accepts forms such as "₹1,405.50", "3:10 pm" and "1,200 shares". The stored plan adds the session, the reference price, the time from which it is valid (the decision time, when the plan is made during its session) and any notes.

The checks record notes and never change the plan, with one exception: an exit-by time that is later than the square-off, missing, or not after the open or the decision time is set to the square-off. The other checks are that the direction must agree with the rating (Buy or Overweight long; Hold none; Underweight or Sell short or none), that the stop and target lie on the correct sides of the entry, and that the size does not exceed what the sizing rule allows.

6.4 Settlement on 5-minute bars

Settlement waits until the session's close is final (16:00 IST for NSE and BSE), loads that session's 5-minute bars and replays the plan with assumptions that lean against it:

- A long entry at or above the reference price is a buy stop (a breakout); below it, a limit order.
- A bar fills an entry only when it reaches the price, at the bar's open if it opens beyond it.

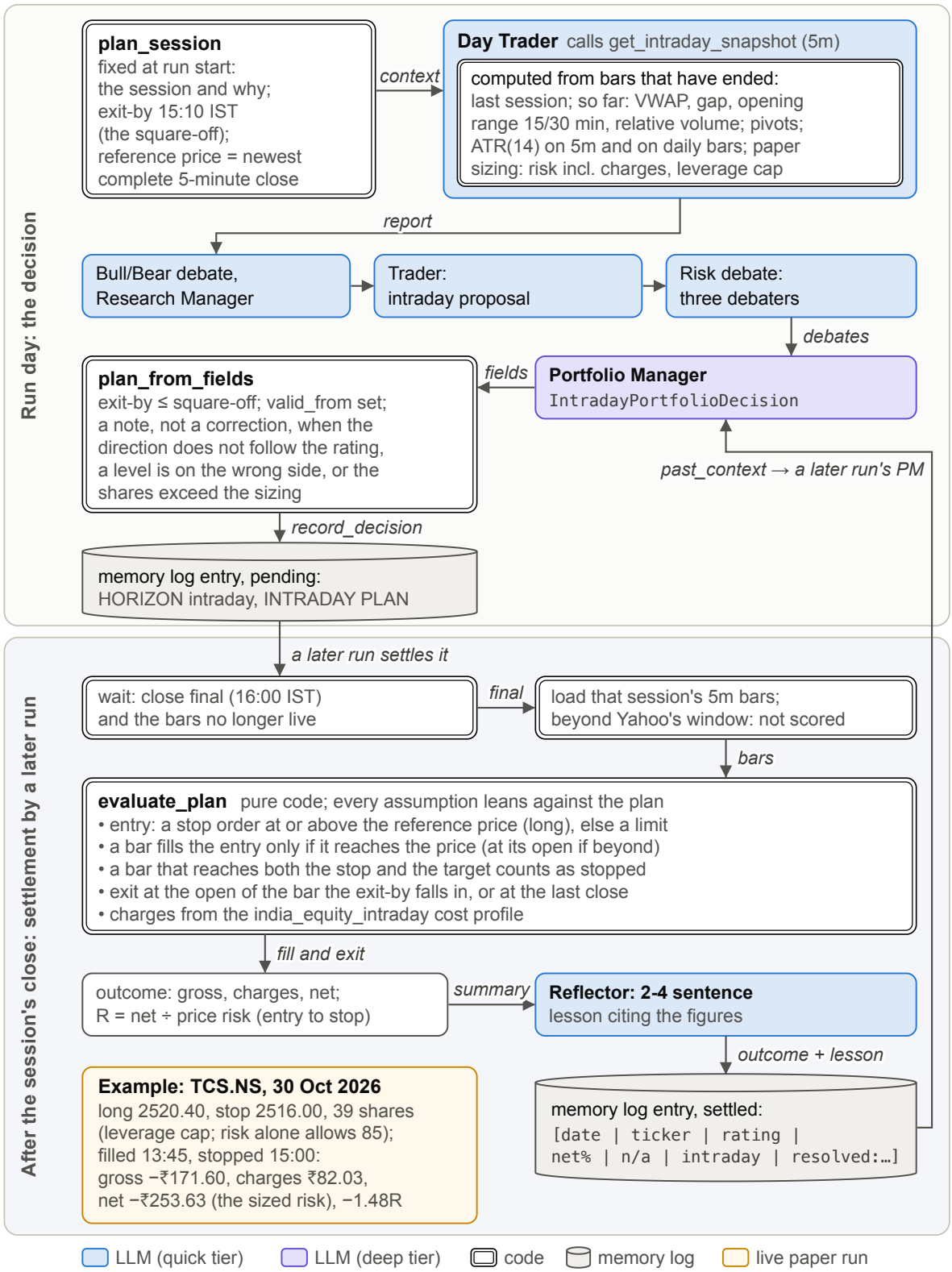


Figure 7. Life of an intraday plan. On the run day, the session context is fixed at the start, the Day Trader reads the intraday snapshot, the debates proceed as in Figure 2, and the Portfolio Manager's plan is checked and stored as data. After the session's close is final, a later run replays the plan on that session's 5-minute bars with charges, and a model then writes a lesson that only future Portfolio Managers read. Inset: the TCS.NS plan of 30 October 2026. Sources: `yahoo/intraday.py`, `markets/intraday.py`, `agents/intraday_plan.py`, `memory/intraday_settlement.py`, `markets/intraday_eval.py`.

- On the fill bar, the stop is checked against the bar's low (or high), but the target only against its close. On later bars, when one bar reaches both the stop and the target, the stop is taken first.
- The position exits at the open of the bar containing the exit-by time, or at the last bar's close.

Day Trader tool get_intraday_snapshot: INFY.NS, 5m

```
# Intraday snapshot for INFY.NS

- Data: Yahoo Finance 5m bars (regular session), through Fri 30 Oct 2026 15:30 IST; sessions in the bars: 26 Oct to 30 Oct 2026 (5).
- Plan for: Mon 02 Nov 2026, the next trading day; today's session closed at 15:30 IST.
- Exit by: 15:10 IST, the configured broker square-off (intraday_square_off): intraday positions are closed the same day.

## Last complete session: Fri 30 Oct 2026 (from the 5m bars)

| Open | High | Low | Close | Volume | Volume vs the sessions before it |
|---:|---:|---:|---:|---:|---:|
| 1149.00 | 1185.00 | 1149.00 | 1185.00 | 12,548,555 | 0.73x |

## The plan's session (Mon 02 Nov 2026) has no completed bar yet
Its open, VWAP, gap, opening range and relative volume do not exist until it trades.

## Classic floor pivots from the 30 Oct 2026 high, low and close

| R3 | R2 | R1 | P | S1 | S2 | S3 |
|---:|---:|---:|---:|---:|---:|---:|
| 1233.00 | 1209.00 | 1197.00 | 1173.00 | 1161.00 | 1137.00 | 1125.00 |

## Volatility

| Measure | Value |
|---|---:|
| ATR(14) on the 5m bars | 2.92492 |
| ATR(14) on daily bars through 30 Oct 2026 | 41.9218 |

## Paper sizing examples (not a recommendation; no order is placed)

| Example | Entry | Stop | Shares | Exposure | Margin | At risk, charges included | Round-trip charges |
|---|---:|---:|---:|---:|---:|---:|---:|
| Long at the newest close, stop 1 x ATR below | 1185.00 | 1182.08 | 84 (leverage cap) | ₹99,540.00 | ₹99,540.00 | ₹328.14 | ₹82.45 |
| Long at the newest close, stop 2 x ATR below | 1185.00 | 1179.15 | 72 | ₹85,320.00 | ₹85,320.00 | ₹498.54 | ₹77.35 |
| Short at the newest close, stop 1 x ATR above | 1185.00 | 1187.92 | 84 (leverage cap) | ₹99,540.00 | ₹99,540.00 | ₹328.23 | ₹82.53 |
| Short at the newest close, stop 2 x ATR above | 1185.00 | 1190.85 | 72 | ₹85,320.00 | ₹85,320.00 | ₹498.69 | ₹77.50 |

Paper capital ₹1,00,000.00, risking 0.5% a trade (₹500.00) with a leverage cap of 1x (margin = exposure / cap). Shares = floor(risk / (stop
distance + round-trip charges per share)), capped at capital x leverage cap / entry. Charges are estimates under the 'india_equity_intraday'
example profile; verify them with your broker.

## Recent 5m bars (last 12, IST; labelled by their start)

| Start | Open | High | Low | Close | Volume |
|---|---:|---:|---:|---:|---:|
| 14:25 | 1179.40 | 1181.40 | 1176.20 | 1176.70 | 721,889 |
| 14:30 | 1176.35 | 1181.90 | 1176.35 | 1181.90 | 79,371 |
| 14:35 | 1181.55 | 1182.00 | 1180.60 | 1181.50 | 99,748 |
| 14:40 | 1181.30 | 1181.90 | 1180.50 | 1181.15 | 76,546 |
| 14:45 | 1181.45 | 1183.65 | 1181.10 | 1183.05 | 168,885 |
| 14:50 | 1183.00 | 1183.90 | 1182.00 | 1182.55 | 205,929 |
| 14:55 | 1182.60 | 1183.20 | 1181.30 | 1182.20 | 210,492 |
| 15:00 | 1182.20 | 1182.50 | 1181.10 | 1181.75 | 302,439 |
| 15:05 | 1181.75 | 1182.25 | 1180.25 | 1180.50 | 506,794 |
| 15:10 | 1180.45 | 1180.55 | 1175.00 | 1176.20 | 418,083 |
| 15:15 | 1175.00 | 1175.00 | 1175.00 | 1175.00 | 152 |
| 15:25 | 1185.00 | 1185.00 | 1185.00 | 1185.00 | 1,239,454 |

## Notes
- Yahoo's intraday bars come through the unofficial yfinance library: regular session only, labelled by their start, not dividend-adjusted; the
official open and close can differ slightly from the first and last bar.
- NSE hours in force on 30 Oct 2026: pre-open 09:00-09:15 IST; stocks with derivatives trade continuously 09:15-15:15 and then in a closing
auction 15:15-15:35 (within 3% of the 15:00-15:15 average price), other stocks 09:15-15:30 closing on the 15:00-15:30 average; post-close
session 15:50-16:00.
- Yahoo carries no pre-open trades, so the first bar can show little or no volume. Intraday (MIS) positions are squared off the same day; short
selling is allowed intraday only.
- NSE is closed (after the close) at Fri 30 Oct 2026, 22:15 IST; regular session 09:15-15:30 IST. The latest complete session is Fri 30 Oct
2026.
- The plan's session has not traded in these bars: the levels come from completed sessions.
- Paper capital: intraday_paper_capital, a configured paper notional, not an account balance.

Use this snapshot as the source of truth for intraday prices, levels, volumes and paper sizes. If another tool's output conflicts with it, flag
the discrepancy rather than inventing a reconciled number, and do not state an intraday price, level or size that is neither here nor in another
tool's output.
```

Figure 8. The intraday snapshot the Day Trader receives for INFY.NS, dated 30 October 2026 (5-minute bars, five sessions), captured at 22:15:15 IST by invoking the tool as the graph does. It plans for Monday 2 November with exit by 15:10 IST, notes that the plan's session has no completed bar yet, and gives floor pivots (R1 1197.00, P 1173.00), ATR(14) of 2.92492 on 5-minute bars and 41.9218 on daily bars, four paper-sizing rows with charges at the default ₹1,00,000 of paper capital, the last 12 bars and notes on NSE hours. Rendered at 144 columns so that no table row wraps. The plan in Section 8.6 was sized against a ₹10,000 book, so its 7 shares differ from the 72 and 84 shares shown here; its stop (1179.15, the two-ATR row) and target (R1) match.

- Bars before the plan's valid-from time are ignored, so a plan made at 13:42 cannot be filled by an earlier bar.

Charges come from the configured intraday profile. The outcome records gross and net profit, charges, net return as a percentage of the position's value, and the R multiple. A plan outside Yahoo's intraday window (about 60 days for 5-minute bars) is marked as not scored.

The R multiple is net profit divided by the money risked, where the money risked is the price distance from entry to stop times the number of shares, without charges. A stop-out exactly at the stop therefore reads below −1R by the ratio of charges to price risk. Swing settlement records an excess return against a benchmark; intraday settlement records none because, as the code puts it, "there is no benchmark for one session".

6.5 Worked example

Table 7 recomputes the TCS.NS plan of Section 8 with the fork's own sizing and charge functions under the default profile. The risk budget is ₹500, 0.5% of ₹1,00,000. Risk alone would allow 85 shares, but the 1× leverage cap allows floor(1,00,000 / 2,520.40) = 39, so the cap binds. At the stop, the money at risk including charges is ₹253.63, exactly the realised net loss. The loss stayed within the budget, and R reads −1.48 only because ₹82.03 of charges sit on top of ₹171.60 of price risk.

Table 7. The TCS.NS paper plan of 30 October 2026 recomputed with the repository's `intraday_position`, `order_costs` and `trade_charges` functions under `india_equity_intraday`. The figures match the settled outcome in the memory log and an independent hand calculation of the charges.

Item	Value
Entry (buy stop) / stop / target	2520.40 / 2516.00 / 2527.73
Paper capital, risk per trade, leverage cap	₹1,00,000; 0.5% (₹500); 1×
Shares allowed by risk alone / by the leverage cap	85 / 39 (binding)
Buy-order charges	₹30.23 (brokerage 20.00, exchange 3.02, SEBI 0.10, stamp 2.95, GST 4.16)
Sell-order charges	₹51.80 (brokerage 20.00, STT 24.53, exchange 3.01, SEBI 0.10, GST 4.16)
Round-trip charges	₹82.03
Gross P&L at the stop (4.40 × 39)	−₹171.60
Net P&L	−₹253.63
Net return on position value (₹98,295.60)	−0.26%
R multiple (net P&L / price risk)	−1.48

The plan's own text was candid about the thin edge. The Portfolio Manager wrote that "with 39 shares the gross reward of about ₹286 drops to about ₹204 after ₹82.03 of round-trip charges, against ₹253.81 at risk including charges, or about 0.8R net".

That ₹253.81 differs slightly from the ₹253.63 in Table 7, and the difference illustrates why validators should recompute figures rather than trust displayed ones. The Day Trader quoted ₹253.81 from the snapshot's sizing example with a stop one ATR below the close. The 5-minute ATR was 4.40466, and the snapshot prints prices to six significant digits, so that stop was displayed as 2516.00. The plan then used the displayed 2516.00, at which the risk is ₹253.63. The difference is 18 paise, but the mechanism, a model copying a rounded display value, is general. This explanation is our inference from the report and the code.

6.6 What the scorer does not see

The scorer reads only the structured fields. Conditions written in the Invalidation field or the summary, such as "don't open the trade" if it has not filled by 14:30, "start with about half", or "no new entry after 13:30", are not enforced. The stored plan that the scorer reads has no invalidation field, and there is no separate entry cut-off setting: an entry can fill at any bar before the one containing the exit-by time. The replay also assumes fills at the stated price, with no slippage within a bar beyond an opening gap. Backtests do not score intraday plans at all.

7 Quality assurance by adversarial multi-agent review

7.1 Method

The fork was written with an AI coding assistant, Claude Code. Its checking was organised as workflows in which several AI agents worked in parallel, each with a defined role and area. Three kinds of workflow were used on 30 October 2026:

- **Hunt.** Each agent took a lens, an area of the code, and reported only findings it had reproduced. The bug hunt used six lenses: India data, intraday edge cases, the headless CLI, Agent-Reach and news, memory and reports, and portfolio and arena.
- **Fix.** Each agent owned an area, fixed the items assigned to it on a copy of the tree and added tests. Fixes were merged centrally and the full suite re-run, also under a "night clock" test plugin, kept outside the repository, that puts every listing's date one day behind the machine's to expose code that assumes the two agree.
- **Review.** Fresh agents tried to break the previous round's fixes and reported only problems they could reproduce. Each review's problems went to the next fix round.

Round 1 was a set of central fixes prompted by the live runs, made at about 15:13. Later rounds were copied into the main tree by a script that ran only if the tree still held exactly the expected earlier state. The test files added in each round are named after it: `test_fixes_*` (round 2), `test_review_*` (2b), `test_review2_*` (2c) and `test_review3_*` (2d). Every agent in these workflows ran the same model, claude-opus-5-5.

7.2 Rounds and statistics

Table 8 lists the eight workflows and Figure 9 plots them. Together they reported 126 reproduced problems and recorded 115 fixes, plus two documentation-only items handled centrally. The 54 agent runs used 15,855,387 tokens and 4,667 tool calls.

Table 8. Quality-assurance workflows on 30 October 2026 (times IST). "Reported" counts reproduced findings or problems, with the agents' own severity labels (high / medium / low). Each fix count is the workflow's own summary; no fix workflow skipped an item or judged one invalid.

#	Workflow	Agents	Time	Reported (H/M/L)	Fixed	Docs only
1	Bug hunt, 6 lenses	6	14:10–14:44	56 (9/18/29)	–	–
2	Round 2 fixes	8	15:15–15:47	–	55	0
3	Review of round 2	8	15:51–16:18	29 (2/7/20)	–	–
4	Round 2b fixes	8	18:41–19:08	–	28	1
5	Review of round 2b	8	19:17–19:55	18 (0/1/17)	–	–
6	Round 2c fixes	6	19:57–20:28	–	17	1
7	Final review (6 reviewers + 1 docs writer)	7	21:00–21:44	23 (0/2/21)	–	–
8	Round 2d fixes	3	21:46–22:19	–	15	0
	Total	54		126 (11/28/87)	115	2

Some rows need a note. Round 2 fixed 54 of the bug hunt's 56 findings, plus a 57th found during the live runs: Polymarket timeouts. The other two bug-hunt findings, both about the verified snapshot (indicators from too little history, and a still-trading row without a label), match the central round-1 fix and its test, although no file records that assignment. Round 2d made 15 fixes for 14 distinct problems from the final review's 23, one problem being handled by two fixers. Of the remaining nine, three have matching changes in the code, and six are documentation items whose status was not checked because the documentation was still being edited.

Two trends are visible. High-severity problems fell from 9 in the bug hunt to 2 in the first review and none in the last two; medium ones went 18, 7, 1 and 2. The total did not fall monotonically, however: the final review, with a different team and a documentation writer, reported 23 problems against 18 in the review before it, 21 of them low. These counts describe the process, not the code's defect density. Each round looked at different changes through different lenses, and the severity labels are the agents' own.

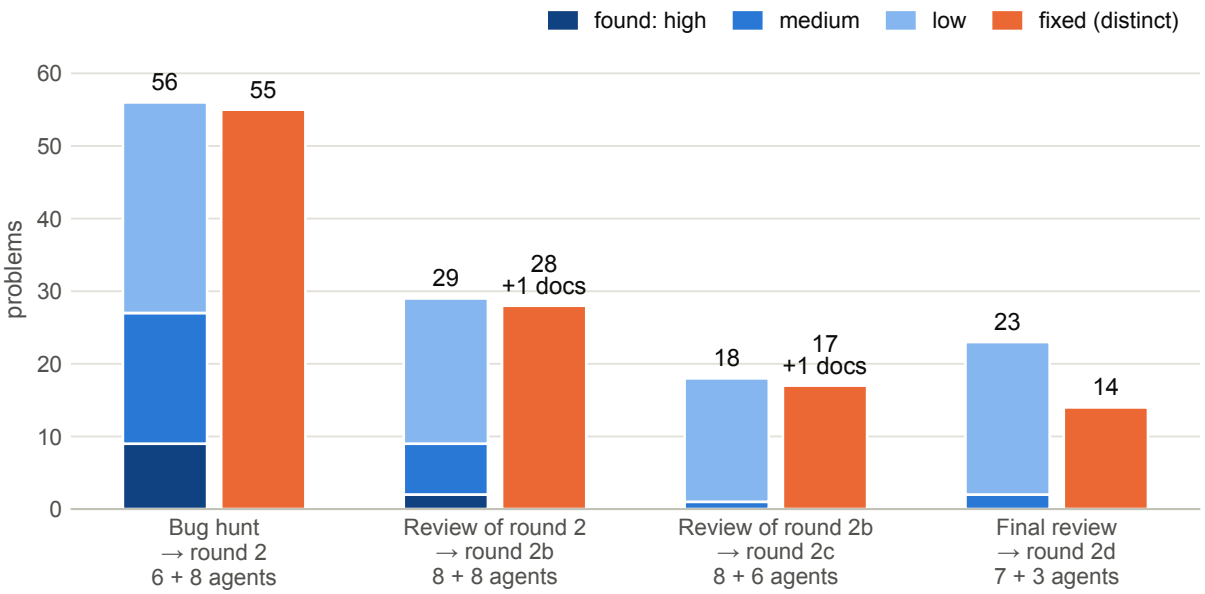


Figure 9. Problems reported by each hunt and review workflow, by severity, and items fixed by each fix workflow, 30 October 2026. Data as in Table 8.

Quality assurance also ran before the first commit. An overnight review by 10 agents on 29–30 October reported 26 findings (1 high, 7 medium, 18 low), all fixed; the integration log records mutation checks of those fixes catching 29 of 29, 20 of 20, 20 of 20 and 10 of 10 mutants. An audit of the commit itself reported 17 findings, of which 6 were confirmed, and a check of the README reported 16, of which 4 were confirmed.

7.3 Test-suite growth

Figure 10 shows the passing test count by milestone. The upstream release passes 1,412 tests. The India work stood at 1,628 on the evening of 29 October; a batch of features (Agent-Reach, the Day Trader, AlphArena and documentation) took it to 2,374; the overnight review to 2,475; and the two pushed commits to 2,490 and 2,501. The uncommitted rounds of 30 October took it to 2,512 (round 1), 2,735 (round 2), 2,926 (round 2b), 3,027 (round 2c), 3,032 (the main tree before round 2d) and 3,139 after round 2d. The upstream release, both commits and the two latest tree states were re-run for this report in a clean environment; the other values come from saved outputs and logs. The final suite also passes under the night clock.

7.4 Mutation checks

A mutation check removes a fix and confirms that its test then fails. On a scratch copy of the current tree, the round-1 script caught 8 of 8 mutants and the snapshot script 2 of 2. A third script, for the round-2b central fixes, defines six mutants but now stops at setup, because rounds 2c and 2d changed the code it patches. Its original results were not saved, so we do not report them. A fourth stopped with a setup error after catching three mutants. Mutation checks show that the tests notice when a fix is removed; they do not show that the fix is right.

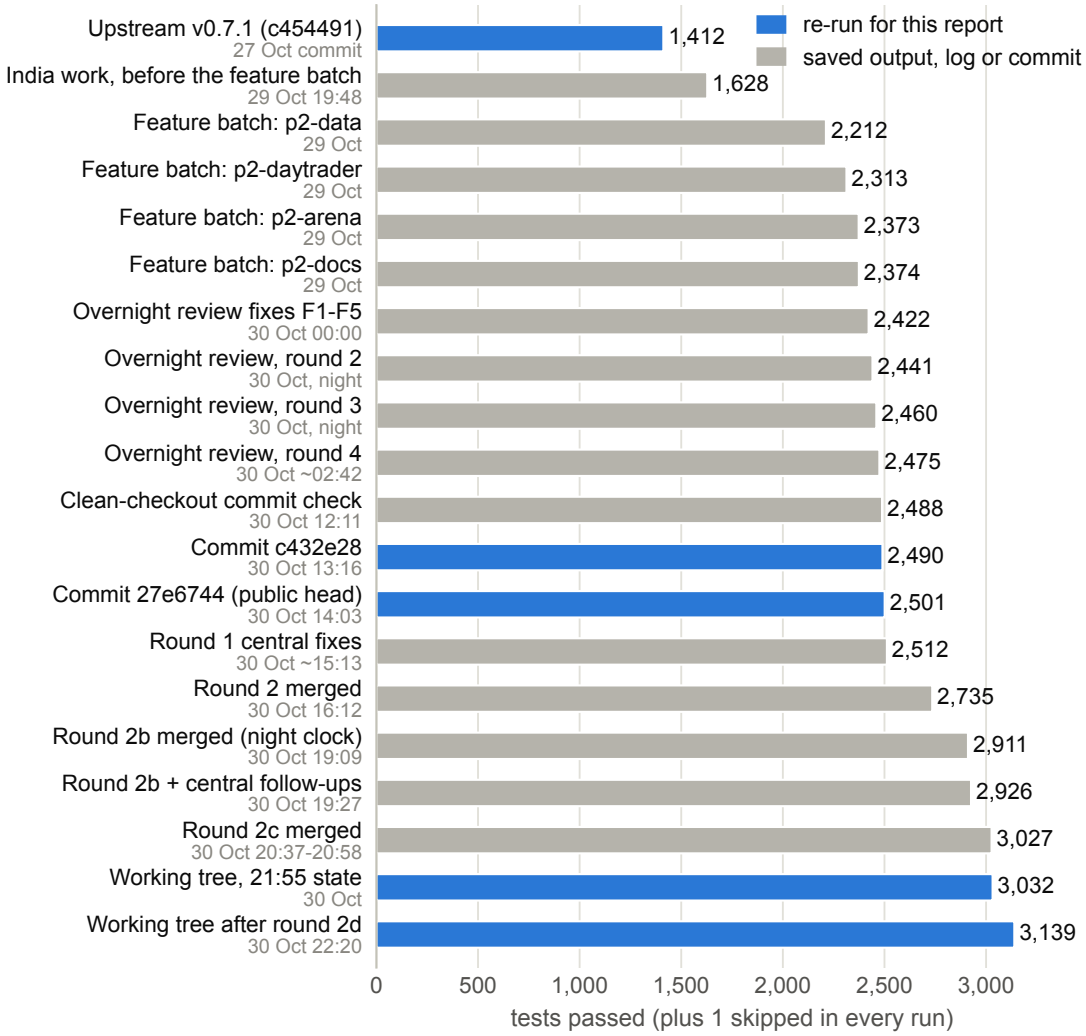


Figure 10. Passing tests by milestone, from upstream v0.7.1 (1,412) through the pushed commits `c432e28` (2,490) and `27e6744` (2,501) to the main working tree after round 2d (3,139). Each run also had one skipped test (an optional dependency) and, where recorded, one deselected integration test. Upstream, both commits and the two latest tree states were re-run on 30 October 2026 between 22:22 and 22:37 IST; the other values come from saved outputs and logs.

7.5 Worked example: a flaky test

In 29 saved full-suite runs during round 2c, one failed. The failing test was `test_polymarket_is_asked_again_after_a_failed_tls_handshake`, which checks the connections the Polymarket client makes. The final review rated this a medium problem and traced it: "The extra connection in the Polymarket tests comes from a post-screen worker that outlives its test".

The diagnosis used a watcher that recorded, at the start of each test, whether a worker thread was alive. Without the fix, its log holds 768 lines, each a test start with the thread `ThreadPoolExecutor-5_0` alive: 32 distinct tests in each of 24 runs. The worker belonged to the sentiment post-screening code, which can call an optional external screening service. A test fixture started it and did not wait for it, so it could still be running, and connecting, when a later test began.

The fix has two parts. The fixture now joins the threads it started, and the test configuration blanks the screening service's key and URL so that no test can reach it. Re-run 25 times on the current tree, no run left a worker alive at a later test's start. The same 25-run loop on a copy without the fix leaked in 15 runs, at 480 test starts. The leak is intermittent, which is why the failure appeared once in 29 runs.

The example shows what the multi-agent review did well here: a rare failure was reproduced, traced to a cause outside the failing test, and fixed at that cause in the test harness. It also shows the limits of a pass count: a suite can pass most of the time while a defect in its own harness waits for the wrong thread scheduling.

7.6 Limits of the method

The agents that wrote, reviewed and fixed the code all ran the same model, so their blind spots are likely to be correlated. Findings count only what agents reproduced, and severities are their judgement. Tests written by the same process can encode its misunderstandings. The author directed the rounds, but the finding, fixing and reviewing were done by agents, and we have no measure of residual defects. The statistics are a record of effort and of the convergence of reported severity, not an estimate of correctness.

8 Case study: live paper runs on 30 October 2026

8.1 Setup

On Friday 30 October 2026 the fork was run live on five NSE stocks and one BSE listing (Table 9). No orders were placed. The runs wrote plans and ratings to a scratch memory log, separate from the author's real log. All nine runs used the `claude_code` provider with claude-opus-5-5 as the deep model and claude-sonnet-5 as the quick model, one research and one risk debate round, and Yahoo Finance data.

The three intraday runs for 30 October used the market, news and Day Trader analysts through the `/day-trader` skill's script, which runs the CLI without a terminal, and sized plans against the default ₹1,00,000 of paper capital. The three swing runs used the Python API with the market, news and fundamentals analysts and a ₹10,000 paper book in rupees. The three plans for Monday 2 November also used the Python API, with the intraday analysts and a ₹10,000 paper book.

Table 9. The nine live paper runs of 30 October 2026, identified by the time each report was generated. Run windows come from the run logs. "Settled" entries were scored by the software after the close; pending swing ratings are scored over five trading days, and the Monday plans after the 2 November session. No orders were placed.

Report	Listing	Horizon	Run window	Rating	Plan or action	Confidence	Status
03:12:49	RELIANCE.NS	Intraday, 30 Oct	03:03–03:13	Underweight	Stand aside	low	Settled: no trade planned
13:42:19	TCS.NS	Intraday, 30 Oct	13:32–13:42	Overweight	Long 39 at 2520.40	low	Settled: stopped out, net −₹253.63
14:16:55	RELIANCE.BO	Intraday, 30 Oct	14:09–14:17	Underweight	Stand aside	low	Settled: no trade planned
14:42:17	HDFCBANK.NS	Swing	14:32–14:44	Underweight	Stay in cash	low	Pending
14:43:03	SBIN.NS	Swing	14:32–14:44	Overweight	Starter of 1 share	low	Pending
14:44:00	TCS.NS	Swing	14:32–14:44	Hold	No purchase	medium	Pending
16:16:13	SBIN.NS	Intraday, 2 Nov	16:08–16:18	Hold	Stand aside	low	Pending
16:18:05	INFY.NS	Intraday, 2 Nov	16:08–16:18	Overweight	Long 7, buy stop 1,185.00	low	Pending
16:18:08	HDFCBANK.NS	Intraday, 2 Nov	16:08–16:18	Underweight	Stand aside	low	Pending

8.2 The TCS.NS intraday plan

The TCS.NS run started at 13:32:04, analysed a 5-minute snapshot with data through 13:30, and finished at 13:42:19 (Figure 11). At the cut-off, the snapshot put TCS above its session VWAP of 2512.71, with relative volume of 0.85. The Portfolio Manager rated the stock Overweight with low confidence and wrote a long plan (Figure 12): a buy stop at 2520.40, a stop at 2516.00, a target of 2527.73 (the R2 pivot), flat by 15:10, and 39 shares. The summary called the size a ceiling:

> start with about half and add the rest only if price breaks above 2527.73 on clearly higher volume. If the entry hasn't filled by 14:30 IST, don't open the trade

The Invalidation field added that the plan should be cancelled or exited if price lost both VWAP and the opening-range low of 2500.80.



Figure 11. Top of the TCS.NS intraday report (run `TCS.NS_20261030_134219`), generated at 13:42 IST on 30 October 2026, before the outcome was known: rating Overweight; NSE, rupees, IST and benchmark NIFTY 50; models `claude_code` (deep claude-opus-5-5, quick claude-sonnet-5); analysts market, news and day_trader; horizon "intraday paper plan", exit by 15:10 IST. The masthead logo and footer credit are those of the upstream project (Tauric Research), which the fork's report template keeps; the fork is not a Tauric product. Screenshot of the saved HTML page, rendered in its light colour scheme.

Plan Session: Fri 30 Oct 2026, the rest of today's session, open until 15:30 IST

Direction: long

Entry Price: 2520.4

Stop Loss: 2516.0

Target Price: 2527.73

Exit By: 15:10

Position Size (shares): 39

Invalidation: Cancel or exit if price trades below VWAP 2512.71 and loses the opening-range low 2500.80 (don't flip short), if no entry has filled by 14:30 IST, or if the position is still open at 15:10 IST.

Time Horizon: Intraday, rest of the 30 Oct 2026 session (flat by 15:10 IST)

Bull Case: The higher-low structure holds and the sector's strength continues: price clears the session high of 2525.00 and reaches 2527.73. If volume picks up, add the second half; 2546.97 (pivot R3) stays out of the plan.

Base Case: Price drifts between VWAP and about 2527, so T1 is hit or the trade is barely positive or flat by 15:10. This fits the light 0.85x volume and the cooling daily momentum (RSI down from 76.51, MACD histogram shrinking over six sessions).

Bear Case: Global AI-trade sentiment turns again, as it did on 27-28 Oct. Price loses VWAP and the 4.40-point stop is hit, possibly with slippage on thin volume.

Key Risks: The whole sector can move 3-5% on outside AI headlines, and light volume plus charges leave a thin edge (about 0.8R net). Not available to the analysis: FII/DII flows, F&O open interest, the stock's price band, exchange announcements, any interim-dividend ex-date, and valuation (fundamentals and sentiment reports were unavailable). Today's bar is still forming, and the two data sources disagree slightly on today's low and open.

Confidence: low

Research output generated by AI agents from delayed or end-of-day market data; it is not investment advice, a prediction or a guarantee. This tool runs in research and paper-trading mode only and places no real orders. Verify every figure with an official source before acting. It is not research or advice from a SEBI-registered research analyst or investment adviser. An intraday plan is a paper exercise: it is not an order and is never sent to a broker.

Figure 12. Plan fields of the Portfolio Manager decision in the same report: plan session, direction long, entry 2520.4, stop 2516.0, target 2527.73, exit by 15:10, 39 shares, the invalidation text, time horizon, bull, base and bear cases, key risks, confidence low, and the page's disclaimer (not investment advice; paper mode; not a SEBI-registered research analyst or investment adviser; never sent to a broker). The full section is in Figure 20.

8.3 Settlement of the TCS.NS plan

The plan became valid at 13:42:19, so the first bar it could use was 13:45. That bar (open 2519.20, high 2521.80) reached the buy stop, and the plan filled at 2520.40. Price rose to a session high of 2527.20 in the 14:15 bar, 0.53 short of the target, and then fell. The 15:00 bar's low of 2515.80 hit the stop, and the position closed at 2516.00. TCS closed the day at 2502.60, after a session low of 2494.70 at 10:35 (Figure 13, Table 10).

The software settled the plan after the close, during the 16:08–16:18 runs, whose Memory Log nodes settle pending decisions as they go. The outcome (Figure 14) was a gross loss of ₹171.60, charges of ₹82.03 and a net loss of ₹253.63: −0.26% of the position and −1.48R. An independent replay of the same rules, written separately, gave the same fill, exit and gross result, and a re-run on bars fetched at 22:31 reproduced it.

The model's reflection, written at settlement, drew a lesson:

> Lesson: when a stop-entry long triggers against a target that sits only marginally above the intraday high already printed, treat it as low-probability and either tighten the target to a level already proven reachable or skip the trade if no fresh high has formed by the entry trigger.

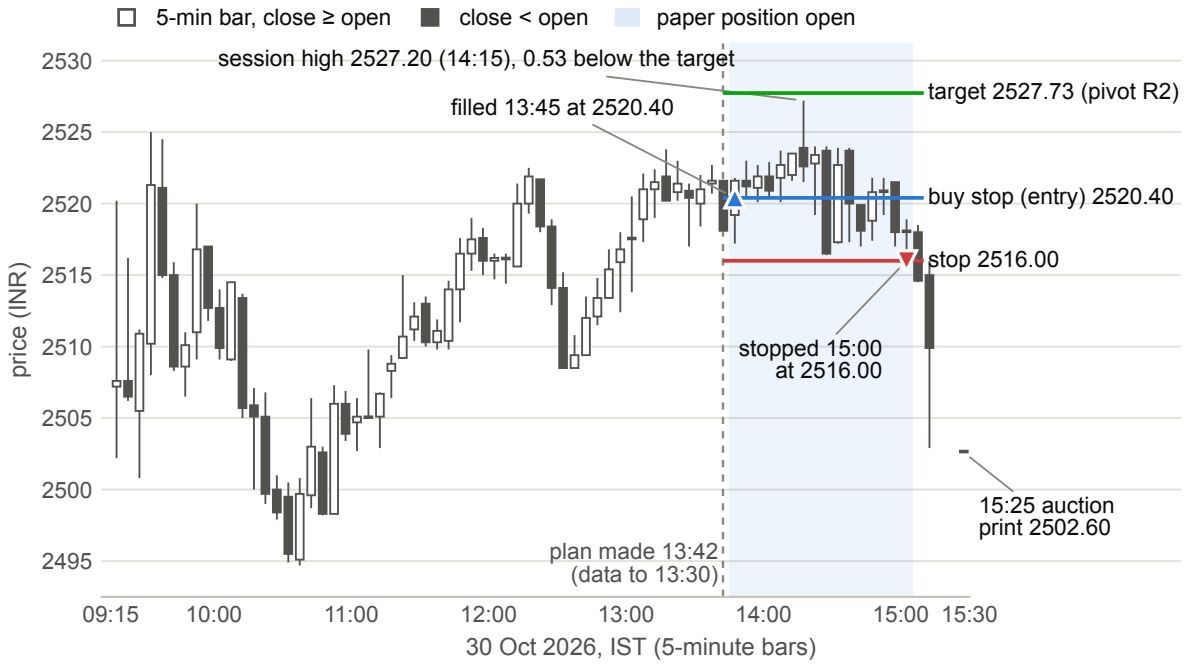


Figure 13. TCS.NS 5-minute bars on 30 October 2026 with the paper plan and its outcome. Levels: buy-stop entry 2520.40, stop 2516.00, target 2527.73 (pivot R2), VWAP at the plan 2512.71 and the 30-minute opening range 2500.80–2525.00. Times: data cut-off 13:30, plan 13:42:19, the plan text's 14:30 no-fill limit and exit-by 15:10. Events: fill at 13:45, session high 2527.20 at 14:15 (0.53 short of the target), stop at 15:00. Bars as served by Yahoo at 22:31 IST: 73 bars from 09:15 to 15:25; Yahoo served no 15:15 or 15:20 bar, and the 15:25 bar is the closing-auction print at 2502.60.

Table 10. Timeline of the TCS.NS paper trade on 30 October 2026, from the 5-minute bars, the report and the memory log.

Time (IST)	Event
09:15–13:30	30-minute opening range (09:15–09:45) 2500.80–2525.00; session low 2494.70 (10:35 bar); high so far 2525.00
13:30	Snapshot cut-off: VWAP 2512.71, relative volume 0.85
13:32:04	Run starts
13:42:19	Plan generated; valid from this time
13:45	Buy stop filled at 2520.40 (bar open 2519.20, high 2521.80)
14:15	Session high 2527.20, 0.53 below the target
14:30	The plan text's "no fill by" time (already filled)
15:00	Stop hit; exit at 2516.00 (bar low 2515.80)
15:10	Exit-by time (position already closed)
15:25	Closing-auction print, 2502.60
After 16:00	Close final; plan settled by the software during the 16:08–16:18 runs

It also noted that "one session, especially one where price missed target by half a point, is too little to indict the setup itself". We agree, and go further: one trade says nothing about whether the system's plans have positive expectancy, and a near miss of 0.53 is no evidence of skill. What the trade does show is that the pipeline works end to end. A plan was frozen with its time stamp; the scorer respected that time, filled and stopped the plan on the bars and applied Indian charges; and the result was reproduced by an independent replay and a hand calculation. Charges added 48% to the gross loss, so the weight of costs that SEBI found in retail intraday trading [24] is visible even in one trade.

Memory log entry (excerpt): TCS.NS intraday plan, settled

```
[2026-10-30 | TCS.NS | Overweight | -0.3% | n/a | intraday | resolved:2026-10-30]

HORIZON: intraday
INTRADAY PLAN: {"direction": "long", "entry_price": 2520.4, "stop_loss": 2516.0, "target_price": 2527.73,
"exit_by": "15:10", "position_size_shares": 39, "invalidation": "Cancel or exit if price trades below VWAP
2512.71 and loses the opening-range low 2500.80 (don't flip short), if no entry has filled by 14:30 IST, or if
the position is still open at 15:10 IST.", "session": "2026-10-30", "reference_price": 2520.39990234375,
"valid_from": "2026-10-30T08:12:19.354769+00:00", "timezone": "Asia/Kolkata", "notes": []}
INTRADAY OUTCOME: {"triggered": true, "entry_time": "2026-10-30T13:45:00+05:30", "entry_price": 2520.4,
"exit_time": "2026-10-30T15:00:00+05:30", "exit_price": 2516.0, "exit_reason": "stop", "gross_pnl":
-171.60000000000355, "charges": 82.02713955760001, "net_pnl": -253.62713955760356, "net_return_pct":
-0.25802491622982465, "r_multiple": -1.4780136337855379, "shares": 39, "entry_type": "stop", "note": "",
"scored": true, "interval": "5m"}

DECISION:
[... 38 lines of the DECISION text omitted in this excerpt (rating, summary, thesis, plan fields, cases,
risks, confidence) ...]

REFLECTION:
The plan filled on the stop entry at 2520.40 (13:45) and was stopped out at 2516.00 (15:00) for a net loss of
253.63 (-1.48R); the session's high of 2527.20 came 0.53 short of the 2527.73 target, and the close at 2502.60
well below entry confirms the long was fighting the day's actual direction. This session undercuts the long
bias and the entry timing (triggering near the session high just before the reversal into the close), while
the stop placement worked cleanly as designed; one session, especially one where price missed target by half a
point, is too little to indict the setup itself, but it's a clear miss on this instance. Lesson: when a
stop-entry long triggers against a target that sits only marginally above the intraday high already printed,
treat it as low-probability and either tighten the target to a level already proven reachable or skip the
trade if no fresh high has formed by the entry trigger.

<!-- ENTRY_END -->
```

Figure 14. Excerpt of the settled TCS.NS entry in the scratch memory log of the live runs, not the author's real log: the header, the stored plan, the settled outcome and the model's reflection. The 38 lines of decision text are elided, as the image states. The outcome JSON prints unrounded values as stored (for example a net P&L of −253.627… and an R multiple of −1.478…). Captured at 22:16:21 IST. The full entry is in Figure 19.

8.4 Stand-asides and the RELIANCE.BO data gap

Two intraday plans for 30 October stood aside. The RELIANCE.NS plan was made at 03:12, before the open, and the RELIANCE.BO plan at 14:16. Both were rated Underweight with direction none and low confidence, and both were settled as "no trade was planned", with no return. The RELIANCE.BO run was the one in which Yahoo served a single daily bar (Section 5.5). A stand-aside is not a successful call: it simply produces nothing to score.

8.5 Swing ratings

Between 14:32 and 14:44, three swing runs used the Python API with a ₹10,000 paper book. TCS.NS was rated Hold with medium confidence ("Don't buy TCS.NS today; we hold no position and keep the ₹10,000 cash"). HDFCBANK.NS was rated Underweight with low confidence ("Underweight means don't open a position today and stay in cash"). SBIN.NS was rated Overweight with low confidence (Figure 15).

The SBIN decision was a starter of one share, about ₹960 or 10% of the book, in a ₹940–965 range before the company's results due on 3 November 2026, paying no more than about ₹990. It set an exit on a daily close below ₹925, a price target of ₹1,031 and a horizon of three to six months. The Portfolio Manager chose to "take 1 share rather than the 2 the 1%-risk rule allows"; the Trader had proposed one to two shares. The thesis cited a trailing P/E of 10.32, a forward P/E of 9.12, a price-to-book ratio of 1.43, a return on equity of 15.18%, and a price below the 50-day (₹989.62) and 200-day (₹1,031.11) averages, and it flagged ₹962.65 as an intraday price. Yahoo's final close that day was ₹961.85 (Figure 4).

These ratings are pending. Swing settlement measures five trading days from the first close after the analysis date, and whatever the outcome, three ratings are not a sample.

V. Portfolio Manager Decision

Portfolio Manager

Rating: Overweight

Executive Summary: Overweight SBIN.NS with staged entry. Buy a starter of 1 share (~₹960, ~10% of the ₹10,000 book) in the ₹940-965 range before Q2FY27 results on 2026-11-03, and don't pay more than ~₹990 (the 50-day average). Exit on a daily close below ₹925. Add more only after results show double-digit NII and net income growth, a credible explanation for the goodwill and derivative jumps, stable NPA figures and no new capital raise; add the last part only after a close back above ~₹990 that holds. Horizon 3-6 months; ₹962.65 is an intraday price on 2026-10-30, so confirm the fill near the close.

Investment Thesis: The case for owning SBIN.NS is stronger than the case for staying out, but the conservative analyst is right on timing, so the position stays small. Supporting evidence (tool data dated 2026-10-30): 10.32x trailing earnings, 9.12x forward, 1.43x book, ROE 15.18%; Q1FY27 net income +13.7% and NII +14.2% YoY; net loans +18.9%; dividend per share up five years running (₹7.10 to ₹17.35). Even after the ₹24,994.53 cr FY2026 capital raise cut diluted EPS growth to 9.9%, the stock looks cheap at about 10x, and no analyst explained why low-to-mid-teens profit growth should trade below ~11x. The aggressive analyst overreached: calling the goodwill (~46x) and derivative-liability (~4x) jumps 'consolidation' is a guess no tool supports; calling the chart a 'confirmed base' is premature (death cross confirmed 2026-09-25, price below the 50-day ₹989.62 and 200-day ₹1,031.11, MACD line -6.60, histogram fading from +4.23 to +2.92); and using 1-year outperformance to dismiss the recent lag (-1.1 points vs NIFTY Bank over 1 month, -1.9 over 3 months) is cherry-picking. Skipping the trade, as the conservative analyst suggests, is too cautious given the valuation, and the plan already makes every later addition conditional on results. The neutral analyst's gap-risk point decides the size: a stop cannot protect against a lower opening after a bad print, so take 1 share rather than the 2 the 1%-risk rule allows, and keep the stop at ₹925 rather than moving it on opinion. Trading costs matter at this size: a 2-share round trip costs ~₹19.62 (~1.02%), and the percentage is likely higher on 1 share (assumption: part of the cost is fixed), against a stop ~3.6% away, so avoid frequent trades. What would change the call: downgrade to Hold or Sell on an EPS miss, a new capital raise, worse NPAs or slippages, goodwill and derivative items left unexplained or shown to be mark-to-market losses, or a close below ₹925. Move toward a full Buy on clean results plus a held reclaim of the 50-day average. Pause adding if the RBI is hawkish at its 2-4 December meeting.

Price Target: 1031.0

Time Horizon: 3-6 months

Bull Case: Q2FY27 confirms double-digit NII and net income growth and the balance-sheet jumps prove benign. A stock at ~10x earnings with a 15% ROE re-rates as it reclaims the 50-day (₹989.62) and then the 200-day (₹1,031.11) average, a gain of about 3-7%, and its 1-year outperformance (+16.8 points vs NIFTY 50) comes back.

Base Case: Results come in roughly in line and the stock trades in a ₹940-990 range while the downtrend slowly heals. The starter position is held and additions wait for a reclaim of the 50-day average; low valuation limits the downside, but the death cross and weak relative strength limit near-term upside.

Bear Case: Results reveal worse asset quality, the goodwill and derivative jumps turn out to come from an acquisition or mark-to-market losses, or another equity raise dilutes shareholders. The stock gaps below the ₹925 stop, the downtrend continues and the multiple shrinks toward ~8x, with a rising India VIX (+14.13% over 1 month) and possible crude and rupee pressure adding to the damage.

Key Risks: Gap risk on the binary 2026-11-03 results that the stop cannot cover; unexplained goodwill (₹66,405.69 cr) and derivative liabilities (₹83,413.87 cr); further dilution; trading costs large relative to a 1-share position. Missing data: no NPA, provision-coverage or slippage figures, no FII/DII flows, F&O open interest or exchange announcements, no verified FX or rates data (the crude/rupee headline is unverified), and today's price is intraday, not a final close.

Confidence: low

Research output generated by AI agents from delayed or end-of-day market data; it is not investment advice, a prediction or a guarantee. This tool runs in research and paper-trading mode only and places no real orders. Verify every figure with an official source before acting. It is not research or advice from a SEBI-registered research analyst or investment adviser.

Figure 15. Portfolio Manager decision in the SBIN.NS swing report (run `SBIN.NS_20261030_144303`), generated at 14:43 IST: rating Overweight; a starter of one share in the ₹940–965 range; exit on a daily close below ₹925; price target 1031.0; horizon 3–6 months; confidence low; and the disclaimer. Its figures (₹962.65 intraday, forward P/E 9.12, 50-day average ₹989.62) are in-session values and differ from the final-close values in Figure 6. The top of the report is in Figure 21.

8.6 Plans for Monday 2 November

After the close, between 16:08 and 16:18, three intraday runs planned for Monday 2 November 2026, each sized against a ₹10,000 paper book. HDFCBANK.NS (Underweight) and SBIN.NS (Hold) stood aside. INFY.NS was rated Overweight with low confidence and given a conditional long:

> Buy stop at ₹1,185.00, triggered only if the first 15–30 min hold above pivot P ₹1,173.00 and Friday's continuous-session low ₹1,175.00. 7 shares (~₹8,295 exposure, ~₹49.74 risk ≈0.5% of paper capital). Stop ₹1,179.15, target ₹1,197.00 (pivot R1, below the 200-SMA at ₹1,200.78). No new entry after 13:30; flat by 15:10 IST.

The ₹49.74 includes charges: $(1{,}185.00 - 1{,}179.15) \times 7 = ₹40.95$ of price risk plus ₹8.79 of round-trip charges, as the Day Trader report states. The Portfolio Manager's bear case says "plus charges", an inconsistency in the model's text. The buy stop equals Friday's closing-auction print, which the decision itself flags as an "auction-set trigger".

The decision cites the TCS trade: "TCS.NS 30-Oct (a late buy stop near the session high was stopped out) leads to a 13:30 entry cutoff and a skip rule if the target is nearly reached at trigger". This is one example of the memory mechanism carrying a lesson forward. It is not evidence that lessons improve decisions.

Two caveats apply before the plan is scored. First, its outcome is unknown at the time of writing; the software will score it after NSE's close becomes final at 16:00 IST on 2 November. Second, the scorer will see only the structured fields (Section 6.6): a buy stop at ₹1,185.00, active from the open because the plan was made before its session, with its stop, target and exit by 15:10. The conditional trigger, the 13:30 cut-off and the skip rule exist only as text and will not be applied. A single outcome, in either direction, will be one more anecdote.

8.7 Defects found by the live runs

The live runs were also a test. They exposed defects that 2,501 passing tests had not caught (Table 11).

Table 11. Defects exposed by the live runs and the corresponding changes in the current code. Counts are from the saved run outputs.

Defect seen live	Evidence	Change
The verified snapshot printed RSI 50.00 and ATR 19.90 from one daily bar (RELIANCE.BO)	Market Analyst report, 14:16 run	Minimum-history rule as in the indicator tool; N/A below it (round 1)
Pandas deprecation notices raised inside `yfinance` filled the output of the `market` command	47 warning lines in the 13:32 output; 0 in the 14:14 output	Notices no longer printed; a test now guards this
A run without a terminal (a scheduled job or the skill) replaced the answers the interactive CLI offers next time; the live scripts had to back up and restore them	Run log	Choices saved only when standard input is a terminal (`cli/selections.py`)
For an intraday plan and a swing rating on the same ticker and date, the second was silently dropped from the memory log, so it was never scored	Code at `27e6744` (one entry per ticker and date); the scratch log now holds both TCS.NS entries for 30 October	One entry per ticker, date and horizon (`memory/log.py`)
Polymarket queries from the author's network timed out after 30 seconds each	32 failure lines across the six Python-API runs	10 s connect and 30 s read timeouts, and a 5-minute pause after a failure (`vendors/polymarket.py`)

Data feeds showed limits too. Agent-Reach's backends answered `RATE_LIMITED` in each of the six Python-API runs (one line in each swing run's error log, and six, seven and four in the Monday runs'), and FRED macro data was unavailable because no key was set.

8.8 What the case study shows and does not show

The case study shows that the India layer, the intraday mode and the settlement path run end to end on a live Indian session with real data; that the scorer's figures can be reproduced independently; that the controls of Section 5 surfaced a real vendor gap; and that live use finds defects that tests miss.

It does not show that the recommendations have value. Over nine runs there was one executed paper trade, and it lost. Three swing ratings and three Monday plans are pending, and the model reported low confidence in eight of the nine decisions. One trading day, five stocks and one model configuration cannot separate skill from chance. The regulatory review in Section 9 would in any case bar presenting such results as performance.

9 Commercial and regulatory readiness in India

9.1 How the assessment was made

The readiness assessment was produced on 30 October 2026 by a nine-agent workflow (18:42–21:03 IST; 3,623,811 tokens; 1,246 tool calls). Five researchers covered SEBI regulation, data protection and cyber security, licences and terms, the code, and the evidence. Three sceptics re-checked each legal or licensing claim against its primary source, and one agent wrote the synthesis.

The sceptics' verdicts were as follows. For SEBI regulation, 147 of 149 claims were confirmed and 2 corrected. For data protection and cyber security, 144 of 148 were confirmed and 4 corrected. For licences and terms, 131 of 152 checks were confirmed, 9 corrected and 12 left unverified. The sceptics added 41 claims of their own. In all, 422 of 449 checks were confirmed, 15 corrected and 12 left unverified. The code and evidence sections were not independently verified, and the assessment labels its code-audit findings and its evidence review accordingly. It states: "Not legal advice: this is the brief for securities counsel and for engineering." We report its verified findings with their claim identifiers; claims that failed verification are not used.

9.2 Bottom line

> The product cannot be sold in India today, to retail or institutional clients, paid or free.

Three reasons support this conclusion. The outputs are regulated research services, and no registration is held. The data sources and two of the model providers are not licensed for commercial use. And, according to the unverified code audit, the system lacks the human review, audit trail and tenant isolation that a regulated service needs. A fourth reason, the absence of evidence that the ratings add value, is the subject of Sections 2 and 11.

The assessment lists 13 stop-ship blockers: no SEBI registration; no human review; unlicensed market data; consumer-plan LLM providers; unlicensed or blocked sources; the shared memory log; no audit trail; performance claims; public paper trading; untrusted text that can steer ratings; missing CERT-In basics; licence notices and branding; and no evidence of value. The memory-log item rests on the unverified code audit, and so do the code findings behind the audit-trail, untrusted-text and CERT-In items; the legal requirements in those items are verified.

9.3 Verified findings that bind a product

Table 12 lists the verified findings that bear most directly on a product built from the fork. Every output of the case study is regulated: the swing ratings and price targets (C002, C003) and the intraday plans (C004). Because an indirect benefit counts as consideration (C005), a free product is no exception.

Institutional buyers cannot fill the gap, since regulated firms may not associate with an unregistered provider, even by connecting systems (C031, C032).

The inputs are not licensed either. The live runs used the `claude_code` provider, which runs on whatever plan the local Claude Code tool is signed in to, and a consumer plan cannot power a product (CC-01). All the prices came from Yahoo through `yfinance`, whose terms bar commercial use (YAH-01), and signals derived from NSE data need NSE's own agreement (NSE-01, NSE-03). Finally, the TCS trade is exactly the kind of result that may not be advertised unless PaRRVA has verified it (C092), and public paper-trading features face SEBI's 2024 statement (SEBI-05). Whether that statement reaches internal or client-only paper tracking is one of the assessment's open questions for counsel.

Table 12. Verified findings from the readiness assessment that bind a product built on the fork, with the assessment's claim identifiers and the sceptics' verdicts. Sources are primary texts as consolidated or fetched on or before 30 October 2026, except where marked.

Area	Finding	Claim	Verdict	Source
Scope	Research services include buy, sell and hold recommendations, price or stop-loss targets, model portfolios and trading calls, whatever the provider's title	C002, C003	Confirmed	[25] reg 2(1)(wa)
Scope	Trading calls are intraday, ultra-short-duration or non-delivery recommendations, or any recommendation that is not personalised	C004	Confirmed	[25] reg 2(1)(zc)
Scope	No one may act or hold out as a research analyst without registration	C006	Confirmed	[25] reg 3(1)
Scope	Consideration is any economic benefit, direct or indirect, from the client or otherwise, so a free tier is not a safe harbour	C005	Confirmed	[25] reg 2(1)(fa)
Scope	Trading calls are not investment advice, so adviser registration cannot carry intraday plans	C013	Confirmed	[28] reg 2(1)(l)
B2B	Regulated persons may not associate with unregistered recommendation providers; association includes IT-system interaction	C031, C032	Confirmed	[27] reg 16A
AI use	A research analyst using AI tools is solely responsible for client-data security, the data used and research based on AI output	C082	Confirmed	[25] reg 24(7)
AI use	The extent of AI use must be disclosed	C083	Confirmed	[25] reg 19(vii)
AI use	From 12 November 2026, a qualified person, licensed where required, must review finance recommendations before delivery	AN-06	Confirmed	[39]
Claims	Advertisements may not refer to past performance or risk-return metrics unless a PaRRVA has verified them	C092	Confirmed	[26] para 11.1(c)(xii)
Claims	No "percentage accuracy" or assured returns, and no statements based on assumptions or projections	C091, C093	Confirmed	[26] para 11.1(c)(x), (ii)
Claims	CARE Ratings is the PaRRVA, with NSE as data centre; regular service from 4 May 2026	C102	Confirmed	[33]
Paper trading	SEBI has stated that apps offering virtual or paper trading on listed companies' prices violate the SCRA 1956 and the SEBI Act 1992	SEBI-05	Confirmed	[34]
Data	Yahoo's terms bar commercial access or reuse, automated collection without permission, and redistribution or derivative works without written permission	YAH-01 to YAH-03	Confirmed	[35]
Data	NSE quotes (real-time) and BSE quotes (15-minute delayed) reach Yahoo from ICE	YAH-06	Confirmed	[36]
Data	Values derived from real-time NSE data for trading decisions are non-display use, which needs an NSE agreement and fee	NSE-01, NSE-03	Confirmed (archived policy; may be superseded)	[37]

Area	Finding	Claim	Verdict	Source
Data	Polymarket is blocked in India	PM-01	Corrected (timing of the block)	[44] (copy hosted by a news site)
Model providers	Products must authenticate with API keys; routing users' requests through Free, Pro or Max plan credentials is not allowed	CC-01	Confirmed	[38]
Model providers	Sign-in with a ChatGPT plan is documented only for open-source and local apps	OAI-04	Confirmed	[40]
Cyber	Listed cyber incidents must be reported to CERT-In within 6 hours	CERTIN-02	Confirmed	[41]
Privacy	The DPDP Act's main duties start 18 months after G.S.R. 843(E), about 13 May 2027	DPDP-04, DPDP-05	Confirmed	[42]
Open source	Modified files must carry change notices when distributed; at 27e6744 the fork's 67 modified files carry none, though its README has a fork notice	AP-03, AP-04	Confirmed; AP-04 corrected	[43]; repository history
Enforcement	In one final order SEBI directed a refund of ₹546.17 crore and held that an educational label cannot mask security-specific calls	C144, C040	Confirmed	[45]

9.4 Recommended path

The assessment considers five routes: registration as a research analyst (as a body corporate), a business-to-business tool, an investment adviser, education only, and a broker partnership. Only research-analyst registration covers all the current outputs (C002–C004), and a body corporate may apply (C041) [25]. Adviser registration cannot carry trading calls (C013) [28]; education cannot include security-specific calls; and a broker partnership is, at the earliest, a third phase. The recommended sequence is:

1. **v1, after registration:** non-personalised swing research on a fixed universe, generated in a pre-market batch, signed by a named analyst and released to all subscribers at once. No intraday plans, quantities, portfolio-aware output or performance claims.
2. **v1.5:** the same research for registered intermediaries, as input to their own analysts, with a diligence pack (sources for each claim, a conflicts page and a model card).
3. **v2:** intraday plans, only after the intraday kill criteria of Section 11 are passed and counsel has ruled on sized plans.
4. **Later:** adviser registration for portfolio-aware advice, and a broker partnership for execution.

Registration requires enlistment with BSE as the research analyst administration and supervisory body before SEBI registration (C049, C050) [25], [26]. It also requires a deposit set by client count, from ₹1 lakh for up to 150 clients to ₹10 lakh above 1,000 (C047) [26]; SEBI fees of ₹20,000 on application, ₹30,000 on grant and ₹5,000 every five years (S-M17) [25]; a principal officer and analysts with a degree or CFA charter plus NISM certification (C042), and a compliance officer (C046) [25]. The assessment names the binding constraints as people, data contracts and a forward, time-stamped track record.

On the engineering side, the assessment keeps the engine (the graph, the structured outputs and `REVIEW` sentinel, the vendor router, the point-in-time guards and the market, cost and replay logic) and puts a review-and-publish workflow around it (Figure 16).

9.5 Implications for the code

The assessment lists code changes that can start before any regulatory decision, together with the fixes it gives for its engineering blockers:

- a commercial build without the consumer-plan providers (`claude_code` and `chatgpt`), the paper-trading service, Polymarket, Reddit, StockTwits, Agent-Reach and the update checker;

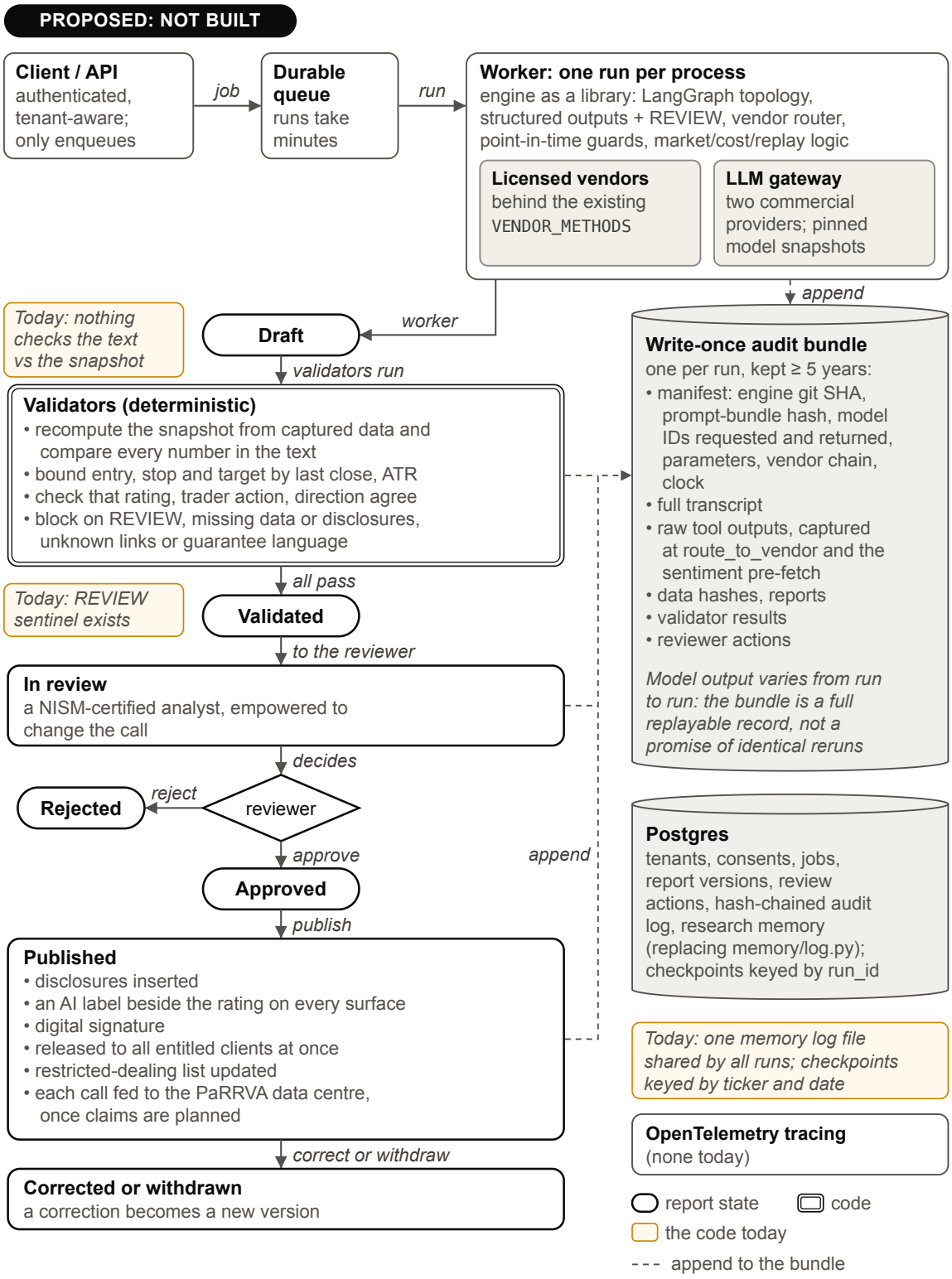


Figure 16. Proposed review-and-publish workflow (not built). Runs are enqueued through an authenticated, tenant-aware API and executed one per worker. Each draft passes deterministic validators: recompute the snapshot from captured data and compare every number in the text; bound entry, stop and target against the last close and ATR; check that rating, trader action and direction agree; and block on `REVIEW`, missing data or disclosures, unknown links or guarantee language. A NISM-certified analyst empowered to change the call then reviews it before publication with disclosures, an AI label and a digital signature. Every step appends to a write-once audit bundle kept for at least five years. Source: the readiness assessment, section 5.

- per-run bundles of prompts, transcripts and raw tool outputs;
- labelling of third-party text, and caps on tool use;
- validators that compare figures with the snapshot;

- settlement as a separate job, and memory scoped to a desk or tenant and holding only reviewed research;
- Apache notices and a new name;
- a lock file, a software bill of materials and a secrets manager.

Several of these address gaps that this report found independently: the absence of any numeric check on swing decisions (Section 5.7), the unenforced text conditions in intraday plans (Section 6.6) and the revision of vendor data after the session (Section 5.6).

10 Limitations and threats to validity

Evidence of value. There is none. One day of live runs, one executed paper trade and a set of pending plans cannot estimate expectancy, and the fork has no forward record yet.

Same-model verification. Every quality-assurance agent ran claude-opus-5-5, and the live runs also used Claude models. Errors to which that model family is prone may pass both writing and review. The severity labels are the agents' own, and the rounds include no separate human code review that we can count.

Measurement of quality assurance. Counts of reported and fixed problems depend on scope and lens and are not defect densities. Test counts measure the suite's size, not its adequacy, and mutation checks cover only a few central fixes.

Non-determinism and reproducibility. LLM outputs vary between runs. According to the assessment's unverified code audit, the fork does not store prompts, transcripts or raw tool outputs for each run, so a decision cannot be replayed exactly. Yahoo also revised a TCS bar's volume after the session (Section 5.6).

Data source. Yahoo Finance, through the unofficial `yfinance` library, is the default for almost all data. Its coverage is uneven (one daily bar for RELIANCE.BO), and its terms bar commercial use. Vendor behaviours stated in this report, such as window lengths, the BSE delay and holiday filler rows, come from the code's comments and were not re-verified against the services.

Scoring model. The intraday replay ignores slippage within bars; it uses Yahoo's 5-minute bars, which lacked the 15:15 and 15:20 bars for TCS on 30 October; it does not apply conditions written as text; and its charge profile is a dated example. The R multiple excludes charges from its denominator.

Look-ahead. The guards restrict the data passed through tools, not the model's pre-training knowledge, and the code does not check the model's training cutoff against the run date. Past-dated runs and backtests are diagnostic only. The deep model used in the live runs, claude-opus-5-5, lists a June 2026 training-data cutoff [48], and the fork's default models list April and May 2026 [47]. The live runs are forward-looking by construction: no model in use on 30 October could have seen that afternoon's bars or the Monday session.

Calendar and rule upkeep. The calendar covers 2026 only. From 1 January 2027 the code will treat only weekends as closed until a new calendar is added. The charge rates are dated 29 October 2026.

Regulatory review. The assessment was researched and verified by AI agents against primary sources. It is not legal advice; some of its sources are archived or secondary; 12 licensing checks remain unverified; and its code and evidence sections were not independently verified. Rules change, and several dates in it fall within weeks of this report.

A moving target. The code changed while this report was being prepared (round 2d was applied at 22:20:42). The figures describe that state, and the documentation was still being edited.

11 Future work

11.1 Engineering

- Validators that recompute the snapshot from captured data and compare every number in a decision's text, and that bound entry, stop and target against the last close and ATR.
- Machine-readable invalidation: entry cut-off times and conditional triggers as plan fields that the scorer enforces.
- Per-run bundles of a manifest, transcript, raw tool outputs and data hashes, including the 5-minute bars used at settlement.
- A training-cutoff check, and a look-ahead propensity diagnostic [8] for any past-dated run; point-in-time models [10], [14] as a control arm.
- Licensed vendors behind the existing router interface, and removal of the consumer-plan providers from any commercial build.
- A 2027 calendar and dated updates of the charge profiles.
- Scoring the INFY.NS plan after 2 November and the three swing ratings when they fall due, reported as anecdotes.

11.2 Pre-registered evaluation protocol

The assessment proposes the following protocol. Its aim is evidence that a regulator, a client and a sceptical researcher would each accept.

1. Pre-register the hypotheses, one primary endpoint, the universe, horizon, costs, benchmarks, baselines, tests and kill criteria; hash and time-stamp the document.
2. Use a point-in-time NIFTY 100 or NIFTY 200 universe keyed by ISIN, on licensed prices as of each date.
3. Count only windows after the model's training cutoff. Pin model snapshots; any change that alters outputs starts a new period, which is never pooled with earlier ones.
4. Keep a forward ledger: one record per call, written before the session it applies to and hash-chained, with a daily root time-stamped under RFC 3161 [49] by a certifying authority licensed by India's Controller of Certifying Authorities [50]. Outcomes are appended, never rewritten.
5. Measure results net of Indian charges and slippage (with a sensitivity range of 0–20 basis points) against total-return indices. Baselines: shuffled ratings, always-Buy, 12-1 momentum, an equal-weight universe and a single-call LLM on the same inputs.
6. Primary metric: the rank information coefficient (IC) of rating against forward excess return, or the net long-short spread, with date-clustered errors; $t \geq 3$ for headline claims.
7. Plan for at least 12 months. As an illustration, the assessment estimates that at a mean IC of 0.03 about 45 weekly dates are needed at 100 stocks (22 at 200) for $t = 2$, and 101 (50) for $t = 3$. Intraday plans would need about 400 triggered plans for $t = 2$ at a mean of 0.1R. These figures depend on assumptions about dispersion and are for planning only.

Under SEBI's rules, results from such a protocol could be shown publicly only as PaRRVA-verified metrics (C092, C100) [26], [27].

11.3 Kill criteria

Table 13 lists the assessment's proposed kill criteria. They are meant to be fixed before any data are seen, so that a negative result ends a claim instead of prompting a search for a better metric.

Table 13. Proposed kill criteria from the readiness assessment. The thresholds are to be set in the pre-registration, before any data are seen.

Criterion	Trigger	Consequence
K1 look-ahead	Post-cutoff results materially below pre-cutoff results, or recall probes showing knowledge of outcomes	Pre-cutoff results barred permanently
K2 no signal	The 95% interval of the net primary endpoint includes zero at the pre-registered sample size	Stop selling ratings as excess return; one pre-set extension if $2 \le t < 3$
K3 no multi-agent premium	No advantage over a single-call LLM on the same inputs	Ship the cheaper system or stop
K4–K5 baselines and costs	Losing to momentum or shuffled ratings after costs, or a net long-short spread at or below zero at realistic slippage	The alpha claim ends
K6 instability	More than about 20% of reruns on identical inputs flipping direction	Not specified beyond failing the criterion
K7 intraday	After at least 400 triggered plans, a 95% upper bound on net expectancy at or below zero, drawdown beyond budget, or no edge over random entry	Intraday plans are not offered (the v2 step requires passing this criterion)
K8–K10 integrity	A broken hash chain, more than 5% failed or outcome-correlated data, or an unpinned model alias	The period is void

12 Conclusion

Adapting a multi-agent research framework to Indian markets is mostly unglamorous work: calendars that change by date, charges that decide whether a trade pays, IST boundaries for news, and data that is sometimes missing or revised. The fork puts that work, and its look-ahead controls, into shared layers of code; it turns gaps into explicit signals; and it scores intraday plans deterministically and net of charges.

An adversarial multi-agent process found and fixed many problems and more than doubled the test suite, but it is not a substitute for independent review. The live case study shows the system working end to end, including a losing trade reproduced to the paisa by independent means; it says nothing about whether the recommendations have value.

A verified regulatory review finds that such a system cannot be sold in India today. The path it recommends runs through registration, licensed data, commercial model APIs, a qualified human reviewer for every call, and a pre-registered forward evaluation with kill criteria. That evaluation, not this report, is what could establish value.

Ethics and disclaimer

Not advice. This report is not investment advice and not research from a SEBI-registered research analyst or investment adviser. It does not recommend buying or selling any security; the stocks named are the subjects of a software case study. It is not legal advice, and it has not been peer reviewed.

No real trading. No real orders were placed. The paper runs were a private engineering test of the software, not a service offered to anyone, and no prizes or contests were involved. Quoted decisions and reflections are model output, not measurements.

Data. Market data came from Yahoo Finance through the unofficial `yfinance` library and is shown here only to illustrate the software. As Section 9 notes, Yahoo's terms bar commercial use; the data source would have to be replaced before any commercial use.

Privacy and security. No personal data about third parties was processed. The terminal captures were made in a clean environment with a scratch home folder, cache and logs, and every capture was scanned for secrets, API keys, e-mail addresses and home-directory paths; none were found, so no masking was needed. The author's real memory log and saved CLI preferences were not opened in preparing this report.

Interest. The author developed the system described here.

Acknowledgements

The engineering described here and this report were prepared with an AI assistant, Claude (Anthropic), used through Claude Code. The assistant wrote and revised code, ran the multi-agent quality-assurance and readiness workflows (whose agents all ran claude-opus-5-5), compiled the evidence notes and drafted this text. The author directed the work and is responsible for its content. The fork builds on TradingAgents by Tauric Research (Yijia Xiao, Edward Sun, Di Luo, Wei Wang and contributors), released under the Apache License 2.0 [1], [43].

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All sources were opened on 30 October 2026. Regulatory and licensing sources are those verified in the readiness assessment.

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Appendix A Configuration and commands

A.1 Reproducing the test counts

The test counts in Table 2 and Figure 10 were reproduced in a clean environment, started from a folder outside the repository:

```
env -i PATH=/usr/bin:/bin:/usr/sbin:/sbin \
  HOME=<scratch>/home \
  TRADINGAGENTS_CACHE_DIR=<scratch>/cache \
  TRADINGAGENTS_RESULTS_DIR=<scratch>/results \
  TRADINGAGENTS_MEMORY_LOG_PATH=<scratch>/memory.md \
  TRADINGAGENTS_NO_UPDATE_CHECK=1 \
  PYTHONPATH=<repo> \
  <venv>/bin/python -m pytest <repo>/tests -q -p no:cacheprovider
```

Python 3.13.14 and pytest 9.2.0 were used. On the code of 22:20:42 IST, 30 October 2026, the expected result is 3139 passed, 1 skipped and 1 deselected. The integration test is deselected by the project's pytest options (`-m "not integration"`), and the skipped test needs the optional `langchain_aws` package. The package loads `.env` and `.env.enterprise` files upward from the working directory at import, so the starting folder and its parents must contain no such file. The earlier revisions were tested from `git archive` exports of each commit.

A.2 Commands

Table 14 gives the commands for the main uses of the fork, including those behind the captures in Appendix B.

Table 14. Commands for the main uses of the fork. A bare symbol with `--exchange nse` resolves to its `.NS` listing.

Purpose	Command
Intraday paper plan on NSE	`tradingagents --ticker TCS --exchange nse --horizon intraday --analysts day_trader,market,news --save --no-show --html`
The same from Claude Code	`/day-trader TCS nse`
Swing analysis for a date	`tradingagents --ticker RELIANCE --exchange nse --date 2026-10-28`
Exchange status, market context and watchlist (no model call)	`tradingagents market --exchange nse --watchlist TCS,INFY,SBIN,RELIANCE,HDFCBANK`
Backtest (swing only; its own memory log)	`tradingagents backtest RELIANCE.NS,TCS.NS --start YYYY-MM-DD --end YYYY-MM-DD --every 7`

The Python API, adapted from the repository's example script (`main.py`) to the three analysts of the swing runs:

```
from tradingagents.default_config import DEFAULT_CONFIG
from tradingagents.graph.trading_graph import TradingAgentsGraph

ta = TradingAgentsGraph(selected_analysts=("market", "news", "fundamentals"),
                        config=DEFAULT_CONFIG.copy())
final_state, rating = ta.propagate("RELIANCE.NS", "2026-10-28")
print(rating)  # Buy, Overweight, Hold, Underweight, Sell or REVIEW
```

A.3 Selected settings

Table 15 lists the settings most relevant to Indian and intraday use. Values are coerced to the default's type, and an invalid value stops the program at start-up.

Table 15. Selected settings, their environment variables and their defaults (`default_config.py`). In all, 36 settings can be overridden from the environment.

Environment variable	Setting	Default
`TRADINGAGENTS_LLM_PROVIDER`	Model provider	`openai`
`TRADINGAGENTS_DEEP_THINK_LLM`, `TRADINGAGENTS_QUICK_THINK_LLM`	Deep and quick models	`gpt-6-sol`, `gpt-6-luna` [47]
`TRADINGAGENTS_MAX_DEBATE_ROUNDS`, `TRADINGAGENTS_MAX_RISK_ROUNDS`	Debate depth	1, 1
`TRADINGAGENTS_MAX_TOOL_ROUNDS`	Tool rounds per analyst	20
`TRADINGAGENTS_DEFAULT_EXCHANGE`	Exchange for bare symbols (NSE or BSE)	None (symbol left bare)
`TRADINGAGENTS_BENCHMARK_TICKER`	A run's own benchmark	None (suffix map)
`TRADINGAGENTS_COST_PROFILE`	Delivery charges	`india_equity_delivery`
`TRADINGAGENTS_RISK_PER_TRADE_PCT`, `TRADINGAGENTS_MAX_POSITION_PCT`	Portfolio sizing reference (2 × ATR stop)	1.0%, 20.0%
`TRADINGAGENTS_DECISION_HORIZON`	Swing or intraday	`swing`
`TRADINGAGENTS_INTRADAY_SQUARE_OFF`	Exit-by time for NSE and BSE plans	15:10
`TRADINGAGENTS_INTRADAY_CAPITAL`, `TRADINGAGENTS_INTRADAY_RISK_PCT`, `TRADINGAGENTS_INTRADAY_LEVERAGE_CAP`	Intraday paper sizing	100000.0, 0.5%, 1.0
`TRADINGAGENTS_INTRADAY_COST_PROFILE`	Intraday charges	`india_equity_intraday`
`TRADINGAGENTS_AGENT_REACH`, `TRADINGAGENTS_ARENA`	Opt-in web research and paper-trading service	Off, off
`TRADINGAGENTS_RESULTS_DIR`, `TRADINGAGENTS_CACHE_DIR`, `TRADINGAGENTS_MEMORY_LOG_PATH`	Output, cache and memory-log locations	Under `~/.tradingagents/`
`TRADINGAGENTS_NO_UPDATE_CHECK`	Skip the upstream release and notice check	Unset

Appendix B Screenshot index

All screenshots were captured on 30 October 2026 between 22:14 and 22:23 IST, after the NSE close, from the main working tree in a clean environment with a fresh cache. No model was called and no analysis was run during the captures. Market data came live from Yahoo Finance: 12 downloads in all, plus the library's own cookie and time-zone lookups.

Terminal output was recorded with its colour codes and rendered to SVG with the rich library (version 15.0.0), using local monospace fonts so that rendering made no network request; the PNG versions were made with headless Chrome at twice the natural size. Each capture was passed through a masking script that replaces home-directory paths with ~ and removes e-mail addresses; it made no replacements, because none appeared, and a final scan found no home path, user name, e-mail address or key-like string in any file. Report pages were copied byte for byte from the saved run folders and screenshotted in their light colour scheme. Table 16 indexes the files, and Figures 17–21 show the captures not used in the main text.

Table 16. Screenshot index. Terminal captures exist as SVG, PNG and plain text; report captures as PNG.

File	Figure	Content	Captured (IST)
`term_a_cli_help`	3	Top-level CLI help	22:14:31
`term_b_market_nse`	17	`market --exchange nse`: session status and market context	22:14:43
`term_b2_market_nse_watchlist`	4	The same with a five-stock watchlist	22:15:05
`term_c_intraday_snapshot_infy`	8	Intraday snapshot tool output, INFY.NS	22:15:15
`term_d_verified_snapshot_sbin`	18	Verified snapshot, SBIN.NS, 30 closes	22:15:47
`term_d2_verified_snapshot_sbin_10rows`	6	Verified snapshot, SBIN.NS, 10 closes	22:18:37
`term_e_memory_tcs_settled_excerpt`	14	Settled TCS.NS memory entry, decision text elided	22:16:21
`term_e_memory_tcs_settled_full`	19	Settled TCS.NS memory entry in full	22:16:21
`report_tcs_intraday_top`	11	TCS.NS intraday report, top	22:22–22:23
`report_tcs_intraday_plan_fields`	12	TCS.NS Portfolio Manager decision, plan fields	22:22–22:23
`report_tcs_intraday_pm_decision`	20	TCS.NS Portfolio Manager decision, full section	22:22–22:23
`report_sbin_swing_top`	21	SBIN.NS swing report, top	22:22–22:23
`report_sbin_swing_pm_decision`	15	SBIN.NS Portfolio Manager decision	22:22–22:23

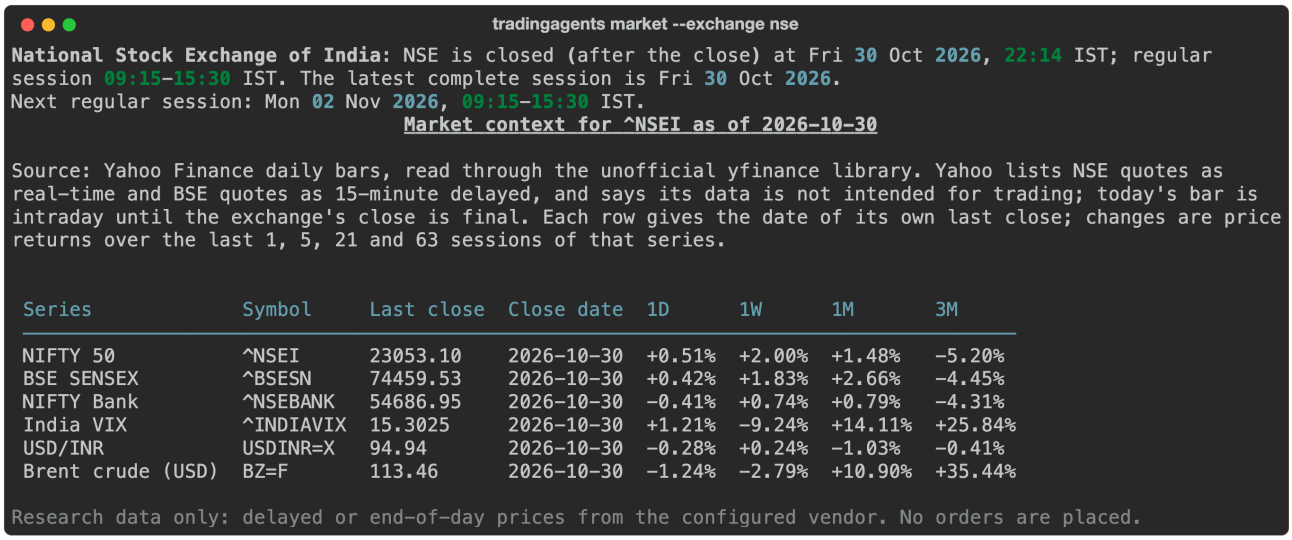


Figure 17. `tradingagents market --exchange nse`, captured at 22:14:43 IST on 30 October 2026: NSE closed (after the close), next session Monday 2 November, and the market context for NIFTY 50 (23053.10), SENSEX (74459.53), NIFTY Bank (54686.95), India VIX (15.3025), USD/INR (94.94) and Brent (113.46), each closing on 30 October, with 1D, 1W, 1M and 3M changes. No model is called.

get_verified_market_snapshot: SBIN.NS

```
## Verified market data snapshot for SBIN.NS

- Requested analysis date: 2026-10-30
- Latest trading row used: 2026-10-30
- Daily rows behind the indicators: 1235, from 2021-11-01
- Rows after the requested analysis date are excluded before verification.
- Prices: adjusted by Yahoo for splits and dividends: rows dated before an ex-dividend date, even one after
the dates asked for, are scaled down by that dividend, so they can sit below the exchange's printed prices and
unadjusted figures such as the intraday bars or a quote's 52-week range.

### Latest verified OHLCV row

| Field | Value |
|---|---:|
| Open | 971.85 |
| High | 971.85 |
| Low | 959.60 |
| Close | 961.85 |
| Volume | 7507038 |

### Verified technical indicators (latest row)

| Indicator | Value |
|---|---:|
| close_10_ema | 961.109 |
| close_50_sma | 989.604 |
| close_200_sma | 1031.00 |
| rsi | 47.3529 |
| boll | 958.06 |
| boll_ub | 975.719 |
| boll_lb | 940.401 |
| macd | -6.66673 |
| macds | -9.53583 |
| macdh | 2.8691 |
| atr | 17.68 |

### Recent verified closes (last 30 rows)

| Date | Close |
|---|---:|
| 2026-09-17 | 988.70 |
| 2026-09-18 | 996.20 |
| 2026-09-21 | 996.00 |
| 2026-09-22 | 987.00 |
| 2026-09-23 | 994.10 |
| 2026-09-24 | 978.50 |
| 2026-09-25 | 983.00 |
| 2026-09-28 | 962.00 |
| 2026-09-29 | 964.70 |
| 2026-09-30 | 959.50 |
| 2026-10-01 | 954.10 |
| 2026-10-05 | 958.00 |
| 2026-10-06 | 958.75 |
| 2026-10-07 | 954.00 |
| 2026-10-08 | 940.00 |
| 2026-10-09 | 959.10 |
| 2026-10-12 | 941.00 |
| 2026-10-13 | 957.00 |
| 2026-10-14 | 954.90 |
| 2026-10-15 | 960.00 |
| 2026-10-16 | 962.60 |
| 2026-10-19 | 973.70 |
| 2026-10-21 | 963.00 |
| 2026-10-22 | 970.00 |
| 2026-10-23 | 958.60 |
| 2026-10-26 | 960.80 |
| 2026-10-27 | 945.10 |
| 2026-10-28 | 956.80 |
| 2026-10-29 | 971.90 |
| 2026-10-30 | 961.85 |

Use this snapshot as the source of truth for exact OHLCV, price-level, and indicator-value claims. If another
tool output conflicts with it, flag the discrepancy rather than inventing a reconciled number. Do not claim
historical validation, support/resistance bounces, or exact percentage moves unless directly supported by tool
output with concrete dates and prices.
```

Figure 18. The verified market snapshot for SBIN.NS with the default 30-day look-back, captured at 22:15:47 IST: latest row 30 October 2026 (open 971.85, high 971.85, low 959.60, close 961.85, volume 7,507,038), the 11 indicators, the last 30 closes and the source-of-truth instruction. Header, prices and indicators are identical to Figure 6.

Memory log entry: TCS.NS intraday plan, settled

```
[2026-10-30 | TCS.NS | Overweight | -0.3% | n/a | intraday | resolved:2026-10-30]

HORIZON: intraday
INTRADAY PLAN: {"direction": "long", "entry_price": 2520.4, "stop_loss": 2516.0, "target_price": 2527.73,
"exit_by": "15:10", "position_size_shares": 39, "invalidation": "Cancel or exit if price trades below VWAP
2512.71 and loses the opening-range low 2500.80 (don't flip short), if no entry has filled by 14:30 IST, or if
the position is still open at 15:10 IST.", "session": "2026-10-30", "reference_price": 2520.39990234375,
"valid_from": "2026-10-30T08:12:19.354769+00:00", "timezone": "Asia/Kolkata", "notes": []}
INTRADAY OUTCOME: {"triggered": true, "entry_time": "2026-10-30T13:45:00+05:30", "entry_price": 2520.4,
"exit_time": "2026-10-30T15:00:00+05:30", "exit_price": 2516.0, "exit_reason": "stop", "gross_pnl":
-171.60000000000355, "charges": 82.02713955760001, "net_pnl": -253.62713955760356, "net_return_pct":
-0.25802491622982465, "r_multiple": -1.4780136337855379, "shares": 39, "entry_type": "stop", "note": "",
"scored": true, "interval": "5m"}

DECISION:
**Rating**: Overweight

**Executive Summary**: Paper long on TCS.NS for the rest of the session. Entry 2520.40, stop 2516.00, target
2527.73 (pivot R2), flat by 15:10 IST. The Day Trader's paper sizing gives 39 shares for this stop, which is
the ceiling: start with about half and add the rest only if price breaks above 2527.73 on clearly higher
volume. If the entry hasn't filled by 14:30 IST, don't open the trade, and don't trade toward the 2546.97
stretch target.

**Investment Thesis**: The trade is long because the intraday setup and the backdrop agree, and the bearish
points concern timing or longer horizons. Intraday 5m snapshot (Yahoo Finance, through 13:30 IST on 30 Oct
2026): TCS.NS gapped up 1.18% from Thursday's close of 2478.00, has stayed above VWAP (2512.71) all session,
and has made higher lows (2508.50 at 12:30, then 2517.00 at 13:25). Market context (30 Oct): NIFTY IT is up
2.24% today and 17.74% over one month, against NIFTY 50 up 0.51% today. Verified daily snapshot (30 Oct): RSI
68.65, MACD 92.46 above its signal at 75.13, price above the 50-day SMA (2255.60) and 200-day SMA (2393.75).
The Research Manager's Overweight call rests on the Q2 FY27 beat (net profit ₹13,884 crore, up about 15%) and
an AI revenue run-rate above $3.1B, up 19.2% quarter on quarter.

Size and conviction stay moderate for three reasons. First, the Conservative and Neutral analysts are right
that relative volume of 0.85x makes this a low-conviction trade, and the plan itself says a light-volume push
to the target is low-conviction. Second, charges eat most of the edge: gross risk is 4.40 points for 7.33 of
reward (about 1.67R), but with 39 shares the gross reward of about ₹286 drops to about ₹204 after ₹82.03 of
round-trip charges, against ₹253.81 at risk including charges, or about 0.8R net (my own arithmetic from the
snapshot figures). The trade only pays if it reaches T1 more often than not. Third, the same AI-rotation
sentiment pulled IT stocks down as much as 5% on 27-28 Oct.

The Conservative analyst's death-cross argument and the research plan's 'don't chase at 2,519.90' both refer
to the multi-month swing horizon and don't break a trade that is flat by 15:10. The Neutral analyst's staged
sizing best answers the valid concerns. What would change the call: losing VWAP together with the
opening-range low, or no fill by 14:30, cancels the trade; a break above 2527.73 on clearly higher volume
justifies adding the second half, still capped at 39 shares.

**Plan Session**: Fri 30 Oct 2026, the rest of today's session, open until 15:30 IST

**Direction**: long

**Entry Price**: 2520.4

**Stop Loss**: 2516.0

**Target Price**: 2527.73

**Exit By**: 15:10

**Position Size (shares)**: 39

**Invalidation**: Cancel or exit if price trades below VWAP 2512.71 and loses the opening-range low 2500.80
(don't flip short), if no entry has filled by 14:30 IST, or if the position is still open at 15:10 IST.

**Time Horizon**: Intraday, rest of the 30 Oct 2026 session (flat by 15:10 IST)

**Bull Case**: The higher-low structure holds and the sector's strength continues: price clears the session
high of 2525.00 and reaches 2527.73. If volume picks up, add the second half; 2546.97 (pivot R3) stays out of
the plan.

**Base Case**: Price drifts between VWAP and about 2527, so T1 is hit or the trade is barely positive or flat
by 15:10. This fits the light 0.85x volume and the cooling daily momentum (RSI down from 76.51, MACD histogram
shrinking over six sessions).

**Bear Case**: Global AI-trade sentiment turns again, as it did on 27-28 Oct. Price loses VWAP and the
4.40-point stop is hit, possibly with slippage on thin volume.

**Key Risks**: The whole sector can move 3-5% on outside AI headlines, and light volume plus charges leave a
thin edge (about 0.8R net). Not available to the analysis: FII/DII flows, F&O open interest, the stock's price
band, exchange announcements, any interim-dividend ex-date, and valuation (fundamentals and sentiment reports
were unavailable). Today's bar is still forming, and the two data sources disagree slightly on today's low and
open.

**Confidence**: low

REFLECTION:
The plan filled on the stop entry at 2520.40 (13:45) and was stopped out at 2516.00 (15:00) for a net loss of
253.63 (-1.48R); the session's high of 2527.20 came 0.53 short of the 2527.73 target, and the close at 2502.60
well below entry confirms the long was fighting the day's actual direction. This session undercuts the long
bias and the entry timing (triggering near the session high just before the reversal into the close), while
the stop placement worked cleanly as designed; one session, especially one where price missed target by half a
point, is too little to indict the setup itself, but it's a clear miss on this instance. Lesson: when a
stop-entry long triggers against a target that sits only marginally above the intraday high already printed,
treat it as low-probability and either tighten the target to a level already proven reachable or skip the
trade if no fresh high has formed by the entry trigger.

<!-- ENTRY_END -->
```

Figure 19. The settled TCS.NS intraday entry in full, from the scratch memory log of the live runs (not the author's real log): header, stored plan, settled outcome, the complete decision text and the reflection. Captured at 22:16:21 IST; long JSON lines wrap.

V. Portfolio Manager Decision

Portfolio Manager

Rating: Overweight

Executive Summary: Paper long on TCS.NS for the rest of the session. Entry 2520.40, stop 2516.00, target 2527.73 (pivot R2), flat by 15:10 IST. The Day Trader's paper sizing gives 39 shares for this stop, which is the ceiling: start with about half and add the rest only if price breaks above 2527.73 on clearly higher volume. If the entry hasn't filled by 14:30 IST, don't open the trade, and don't trade toward the 2546.97 stretch target.

Investment Thesis: The trade is long because the intraday setup and the backdrop agree, and the bearish points concern timing or longer horizons. Intraday 5m snapshot (Yahoo Finance, through 13:30 IST on 30 Oct 2026): TCS.NS gapped up 1.18% from Thursday's close of 2478.00, has stayed above VWAP (2512.71) all session, and has made higher lows (2508.50 at 12:30, then 2517.00 at 13:25). Market context (30 Oct): NIFTY IT is up 2.24% today and 17.74% over one month, against NIFTY 50 up 0.51% today. Verified daily snapshot (30 Oct): RSI 68.65, MACD 92.46 above its signal at 75.13, price above the 50-day SMA (2255.60) and 200-day SMA (2393.75). The Research Manager's Overweight call rests on the Q2 FY27 beat (net profit ₹13,884 crore, up about 15%) and an AI revenue run-rate above $3.1B, up 19.2% quarter on quarter.

Size and conviction stay moderate for three reasons. First, the Conservative and Neutral analysts are right that relative volume of 0.85x makes this a low-conviction trade, and the plan itself says a light-volume push to the target is low-conviction. Second, charges eat most of the edge: gross risk is 4.40 points for 7.33 of reward (about 1.67R), but with 39 shares the gross reward of about ₹286 drops to about ₹204 after ₹82.03 of round-trip charges, against ₹253.81 at risk including charges, or about 0.8R net (my own arithmetic from the snapshot figures). The trade only pays if it reaches T1 more often than not. Third, the same AI-rotation sentiment pulled IT stocks down as much as 5% on 27-28 Oct.

The Conservative analyst's death-cross argument and the research plan's 'don't chase at 2,519.90' both refer to the multi-month swing horizon and don't break a trade that is flat by 15:10. The Neutral analyst's staged sizing best answers the valid concerns. What would change the call: losing VWAP together with the opening-range low, or no fill by 14:30, cancels the trade; a break above 2527.73 on clearly higher volume justifies adding the second half, still capped at 39 shares.

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Exit By: 15:10

Position Size (shares): 39

Invalidation: Cancel or exit if price trades below VWAP 2512.71 and loses the opening-range low 2500.80 (don't flip short), if no entry has filled by 14:30 IST, or if the position is still open at 15:10 IST.

Time Horizon: Intraday, rest of the 30 Oct 2026 session (flat by 15:10 IST)

Bull Case: The higher-low structure holds and the sector's strength continues: price clears the session high of 2525.00 and reaches 2527.73. If volume picks up, add the second half; 2546.97 (pivot R3) stays out of the plan.

Base Case: Price drifts between VWAP and about 2527, so T1 is hit or the trade is barely positive or flat by 15:10. This fits the light 0.85x volume and the cooling daily momentum (RSI down from 76.51, MACD histogram shrinking over six sessions).

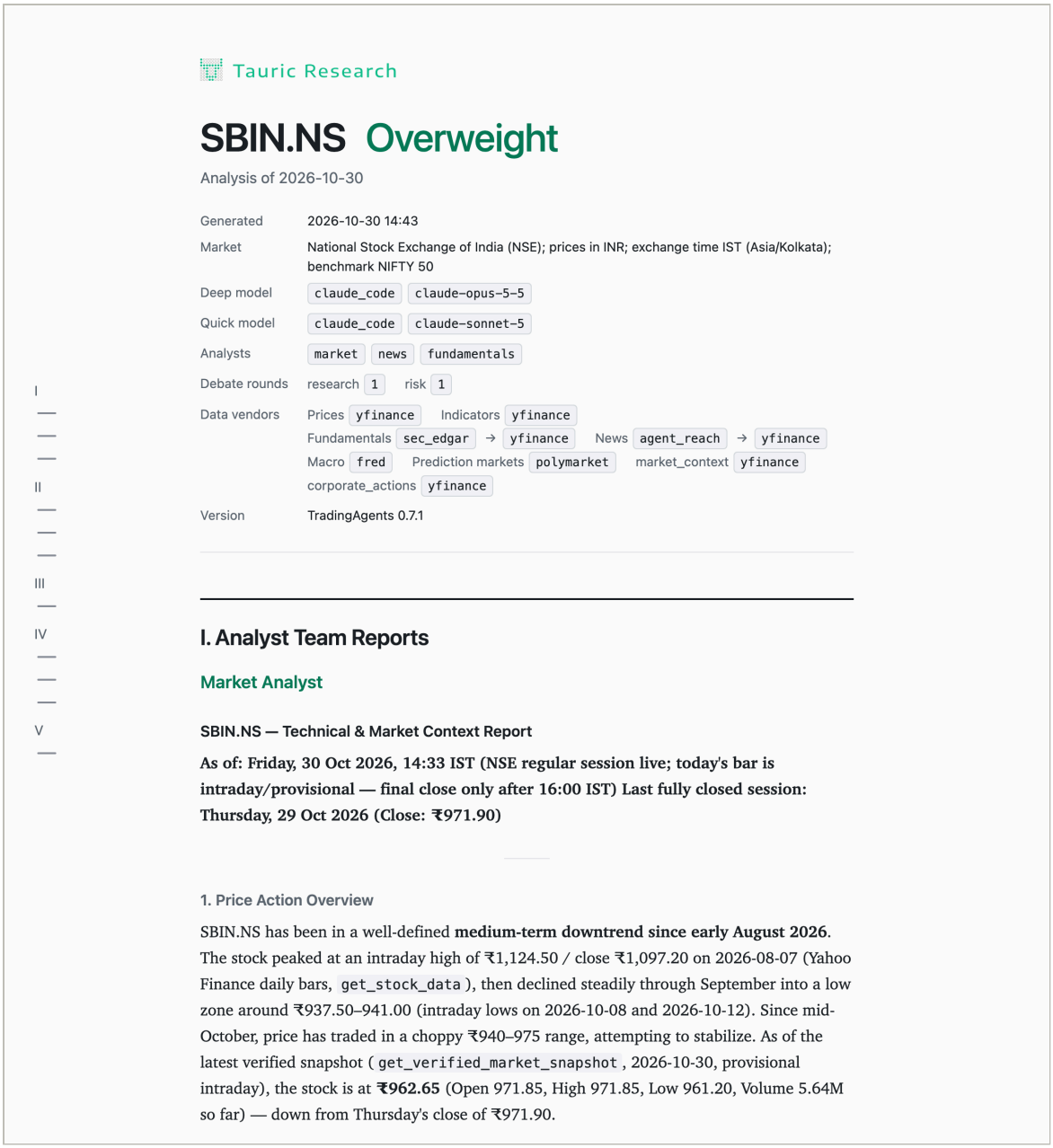
Bear Case: Global AI-trade sentiment turns again, as it did on 27-28 Oct. Price loses VWAP and the 4.40-point stop is hit, possibly with slippage on thin volume.

Key Risks: The whole sector can move 3-5% on outside AI headlines, and light volume plus charges leave a thin edge (about 0.8R net). Not available to the analysis: FII/DII flows, F&O open interest, the stock's price band, exchange announcements, any interim-dividend ex-date, and valuation (fundamentals and sentiment reports were unavailable). Today's bar is still forming, and the two data sources disagree slightly on today's low and open.

Confidence: low

Research output generated by AI agents from delayed or end-of-day market data; it is not investment advice, a prediction or a guarantee. This tool runs in research and paper-trading mode only and places no real orders. Verify every figure with an official source before acting. It is not research or advice from a SEBI-registered research analyst or investment adviser. An intraday plan is a paper exercise: it is not an order and is never sent to a broker.

Figure 20. Section V of the TCS.NS intraday report, the Portfolio Manager decision in full: rating, executive summary, thesis, plan fields, cases, risks, confidence and disclaimer. Cropped from a full-page screenshot of the saved HTML page; generated at 13:42 IST, before the outcome was known.



Tauric Research

SBIN.NS Overweight

Analysis of 2026-10-30

Generated	2026-10-30 14:43
Market	National Stock Exchange of India (NSE); prices in INR; exchange time IST (Asia/Kolkata); benchmark NIFTY 50
Deep model	`claude_code` `claude-opus-5-5`
Quick model	`claude_code` `claude-sonnet-5`
Analysts	`market` `news` `fundamentals`
Debate rounds	research `1` risk `1`
Data vendors	Prices `yfinance` Indicators `yfinance` Fundamentals `sec_edgar` → `yfinance` News `agent_reach` → `yfinance` Macro `fred` Prediction markets `polymarket` market_context `yfinance` corporate_actions `yfinance`
Version	TradingAgents 0.7.1

I II III IV V

I. Analyst Team Reports

Market Analyst

SBIN.NS — Technical & Market Context Report

As of: Friday, 30 Oct 2026, 14:33 IST (NSE regular session live; today's bar is intraday/provisional — final close only after 16:00 IST) Last fully closed session: Thursday, 29 Oct 2026 (Close: ₹971.90)

1. Price Action Overview

SBIN.NS has been in a well-defined **medium-term downtrend since early August 2026**. The stock peaked at an intraday high of ₹1,124.50 / close ₹1,097.20 on 2026-08-07 (Yahoo Finance daily bars, `get_stock_data`), then declined steadily through September into a low zone around ₹937.50–941.00 (intraday lows on 2026-10-08 and 2026-10-12). Since mid-October, price has traded in a choppy ₹940–975 range, attempting to stabilize. As of the latest verified snapshot (`get_verified_market_snapshot`, 2026-10-30, provisional intraday), the stock is at **₹962.65** (Open 971.85, High 971.85, Low 961.20, Volume 5.64M so far) — down from Thursday's close of ₹971.90.

Figure 21. Top of the SBIN.NS swing report (run `SBIN.NS_20261030_144303`), generated at 14:43 IST: rating Overweight, analysis date 30 October 2026, NSE, the same models as Figure 11, analysts market, news and fundamentals, and the start of the Market Analyst's report. As in Figure 11, the masthead is the upstream project's.